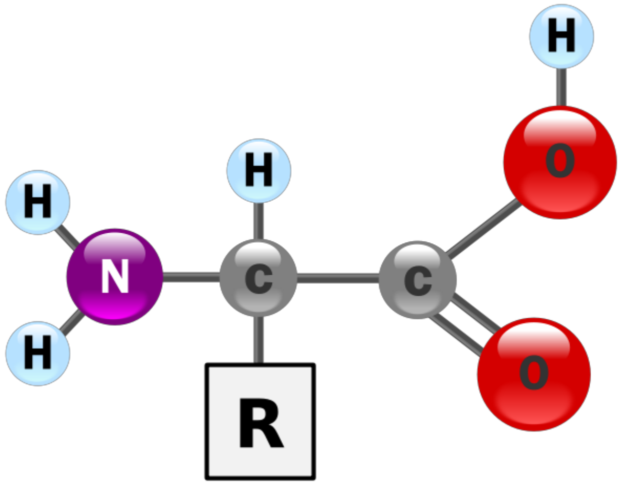
# Amino Acids

Jason Ryan, MD, MPH

# Amino Acids

- Building blocks (monomers) of proteins
- All contain **amine group** and **carboxylic acid**

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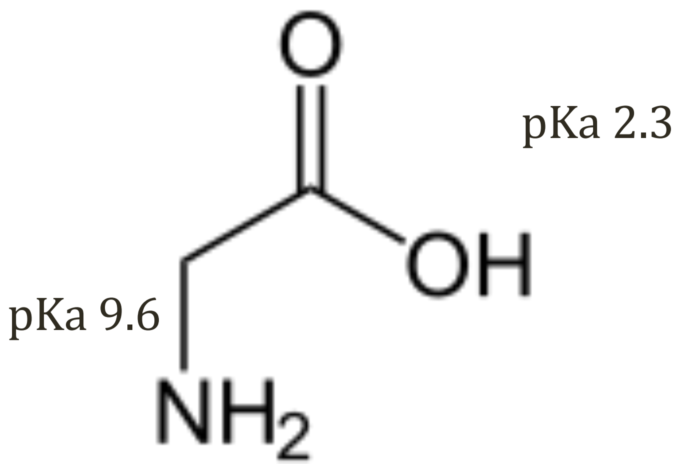
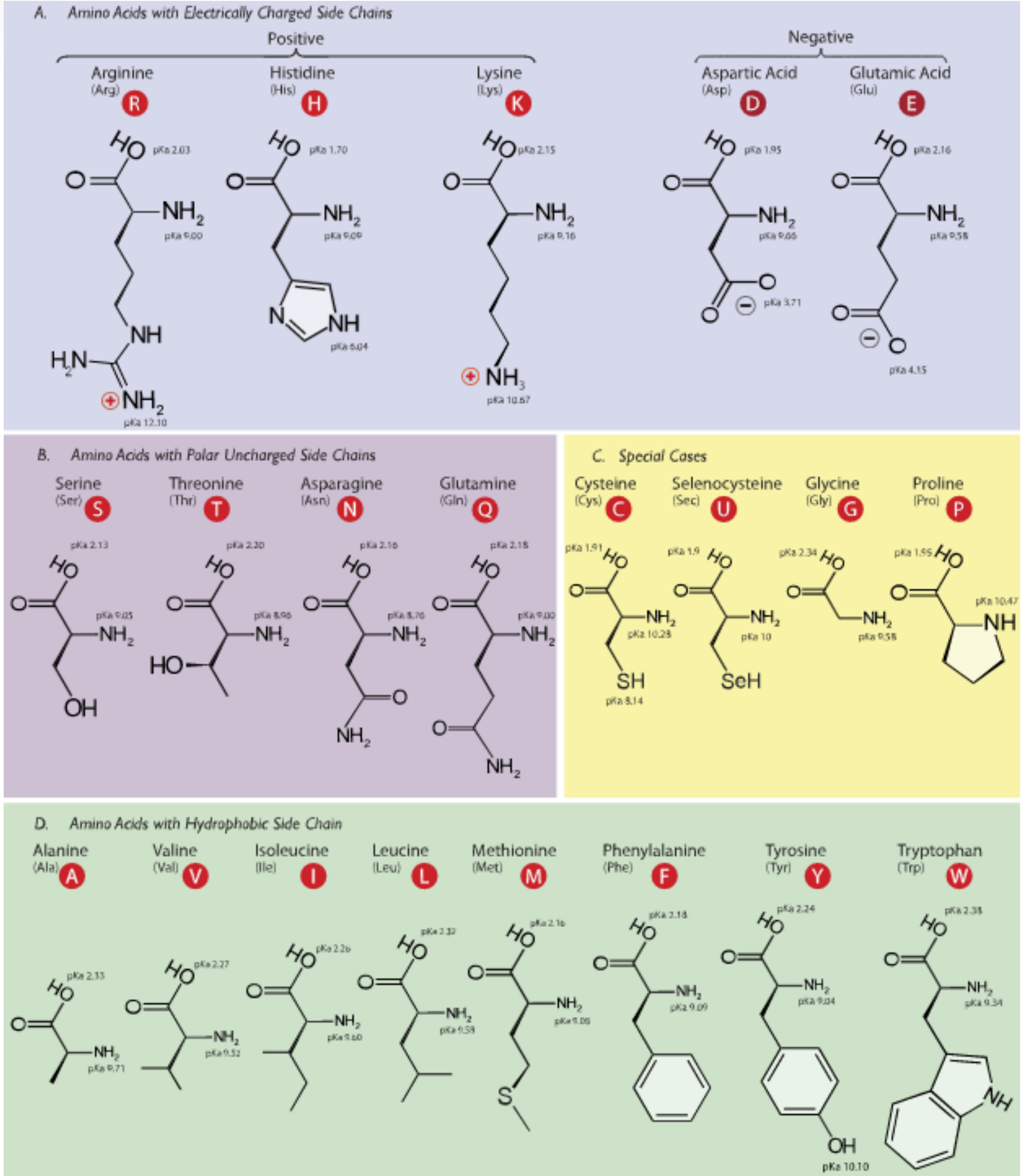


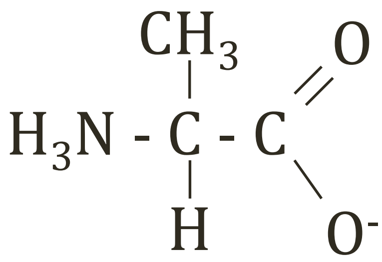

Glycine



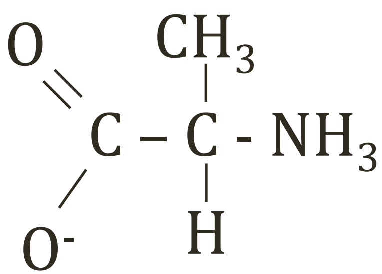
A. Amino Acids with Electrically Charged Side Chains
Positive
Negative
Arginine (Arg) R
Histidine (His) H
Lysine (Lys) K
Aspartic Acid (Asp) D
Glutamic Acid (Glu) E
pKa 2.03
pKa 9.00
pKa 12.10
pKa 1.70
pKa 9.09
pKa 6.04
pKa 2.15
pKa 9.16
pKa 10.67
pKa 1.95
pKa 9.66
pKa 3.71
pKa 2.16
pKa 9.58
pKa 4.15
B. Amino Acids with Polar Uncharged Side Chains
Serine (Ser) S
Threonine (Thr) T
Asparagine (Asn) N
Glutamine (Gln) Q
pKa 2.13
pKa 9.05
pKa 2.20
pKa 8.96
pKa 2.16
pKa 8.76
pKa 2.18
pKa 9.00
C. Special Cases
Cysteine (Cys) C
Selenocysteine (Sec) U
Glycine (Gly) G
Proline (Pro) P
pKa 1.91
pKa 10.28
pKa 8.14
pKa 1.9
pKa 10
pKa 2.34
pKa 9.58
pKa 1.95
pKa 10.47
D. Amino Acids with Hydrophobic Side Chain
Alanine (Ala) A
Valine (Val) V
Isoleucine (Ile) I
Leucine (Leu) L
Methionine (Met) M
Phenylalanine (Phe) F
Tyrosine (Tyr) Y
Tryptophan (Trp) W
pKa 2.33
pKa 9.71
pKa 2.27
pKa 9.52
pKa 2.26
pKa 9.60
pKa 2.32
pKa 9.58
pKa 2.16
pKa 9.08
pKa 2.18
pKa 9.09
pKa 2.24
pKa 9.04
pKa 10.10
pKa 2.38
pKa 9.34

# Amino Acids

- All except glycine have L- and D- configurations
- Only L-amino acids used in human proteins




L - alanine




D - alanine

# pKa

## log acid dissociation constant

$$HA \leftarrow\rightarrow A^{-} + H^{+}$$

$$pH = pKa + \log \frac{[A^{-}]}{[HA]}$$

$$pKa = pH - \log \frac{[A^{-}]}{[HA]}$$

# pKa

$$HA \leftrightarrow A^- + H^+$$

$$pKa: pH - \log \frac{[A^-]}{[HA]}$$

H+ H+

H+ H+

High pH (i.e. 12.0)

$pH > pKa \rightarrow A^- >> AH$

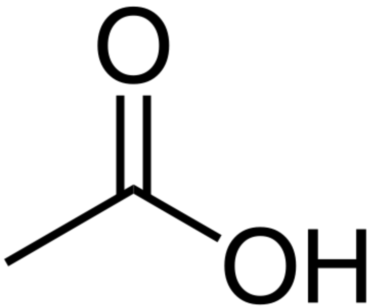
$pH < pKa \rightarrow A^- << AH$

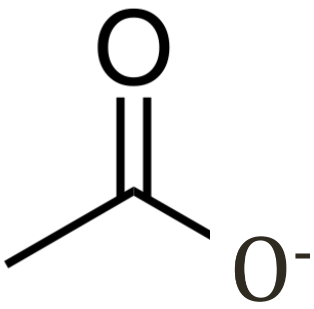
H+ H+
H+ H+ H+
H+ H+
H+ H+ H+
H+
H+ H+ H+
H+ H+ H+

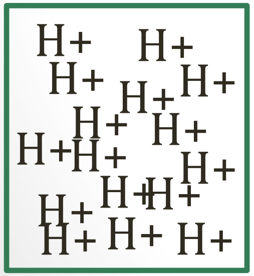
Low pH (i.e. 1.0)

# pKa

- Acetic acid ($C_2O_2H$) pKa = 4.75

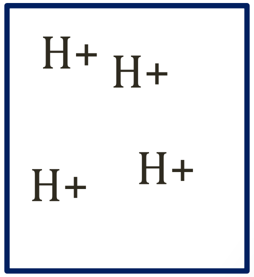


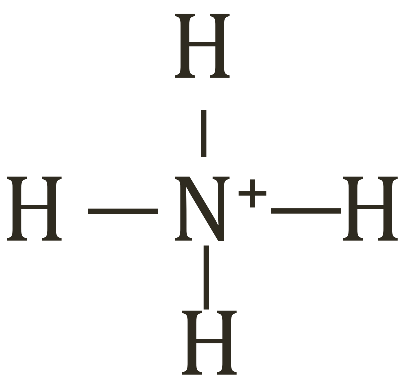

pH <4.75

pH >4.75

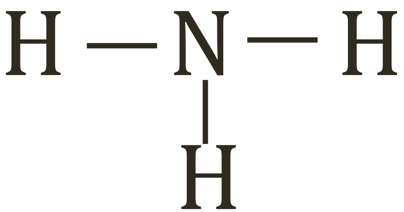


# pKa

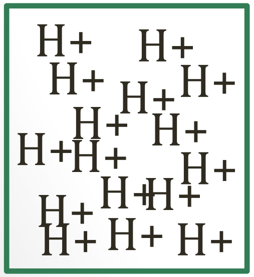
- Ammonia ($NH_3$) pKa = 9.4

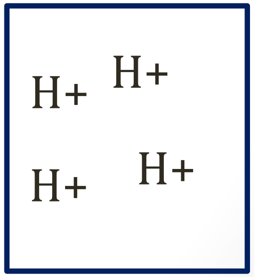


pH <9.4
($NH_4^+$)



pH >9.4
($NH_3$)



# pKa

- Amino acids: multiple acid-base regions
- Each has different pKa

$$\begin{array}{c} R \\ | \\ COOH \end{array} \rightarrow \begin{array}{c} R \\ | \\ COO^- \end{array} + H^+$$

Usually low pKa < 4.0
At normal pH (7.4): COO-

$$\begin{array}{c} R \\ | \\ NH_3^+ \end{array} \rightarrow \begin{array}{c} R \\ | \\ NH_2 \end{array} + H^+$$

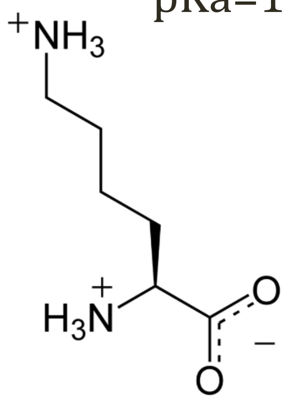
Usually high pKa > 9.0
At normal pH (7.4): $NH_3$

# pKa

- Some side chains have pKa (3 pKa values!)

pKa=10.53

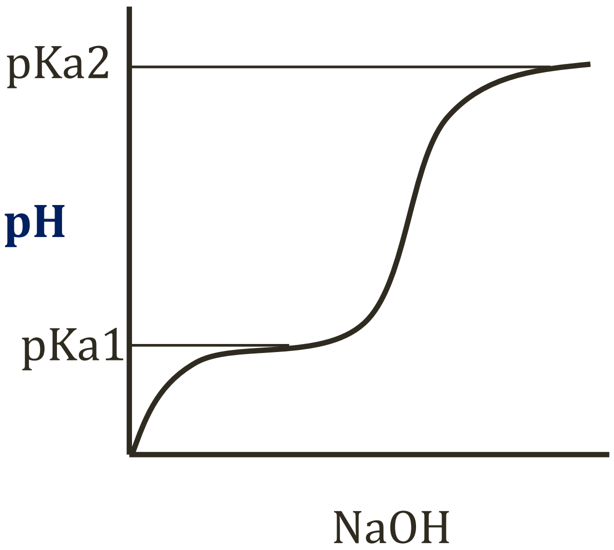
pKa=8.95

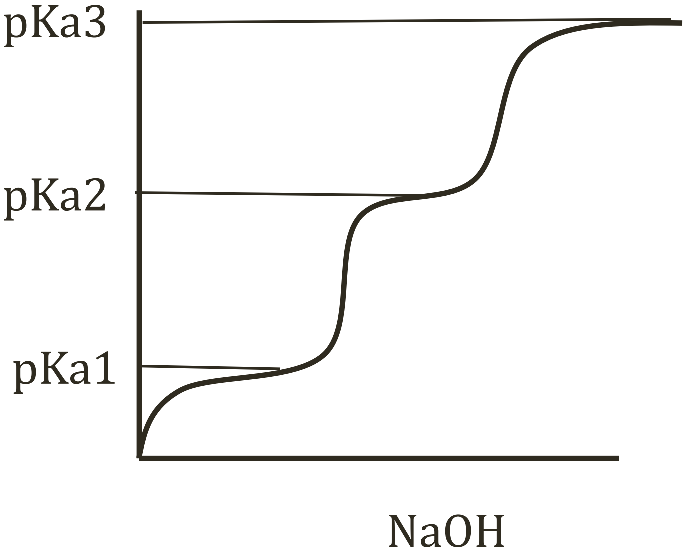


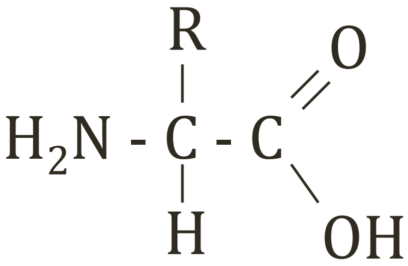
pKa=2.18

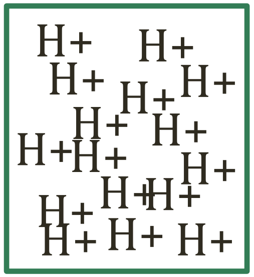
Lysine

# Titration Curves



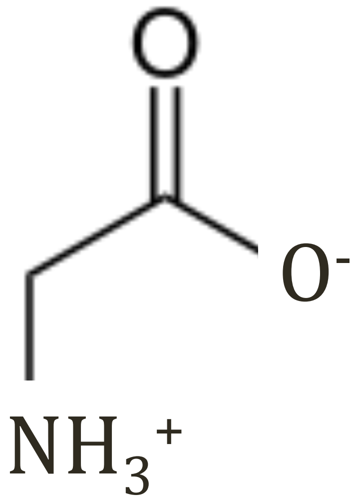



# Charge at Normal pH

- Normal plasma pH=7.4
- AA charge (+/-) depends on pKa values

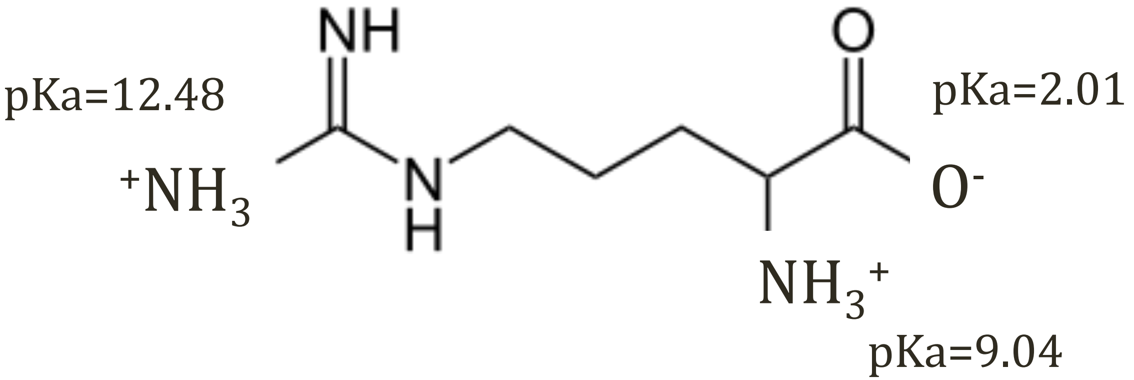



pKa=2.2

pKa=9.6

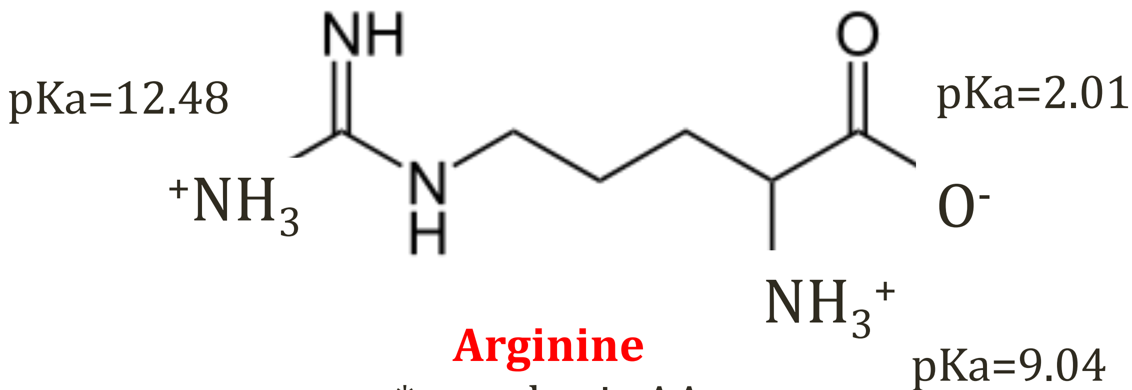
Glycine

# Charge at Normal pH

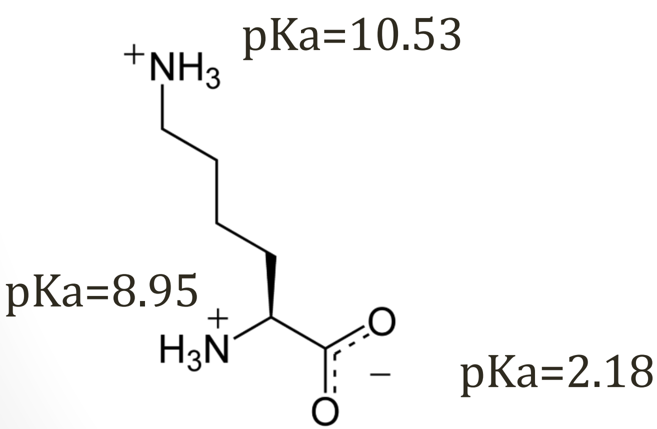



Arginine

# Basic Amino Acids




**Arginine**
*most basic AA



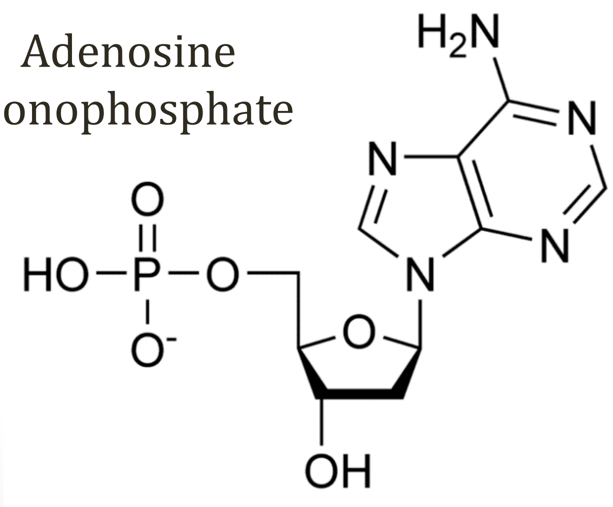

**Lysine**

Both +1 charge at normal pH
Remove $1H^+$ from solution
Raise pH (basic)

# Histones

- Contain **basic** amino acids
  - High content of lysine, arginine
  - Positively charged
  - Binds negatively charged phosphate backbone DNA

Adenosine Monophosphate

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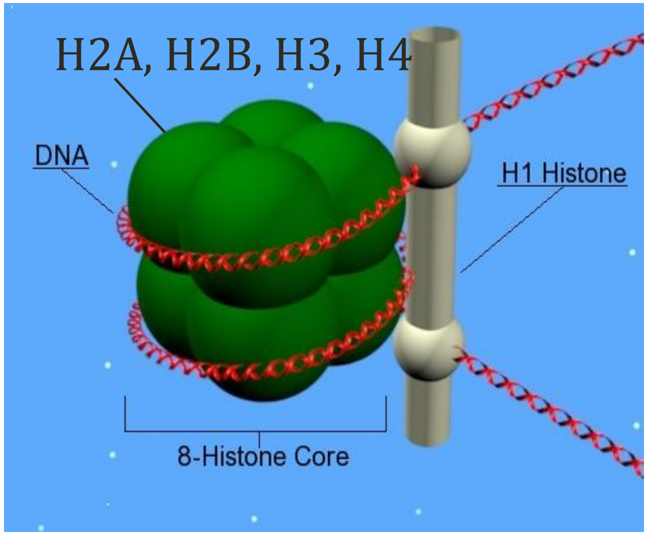


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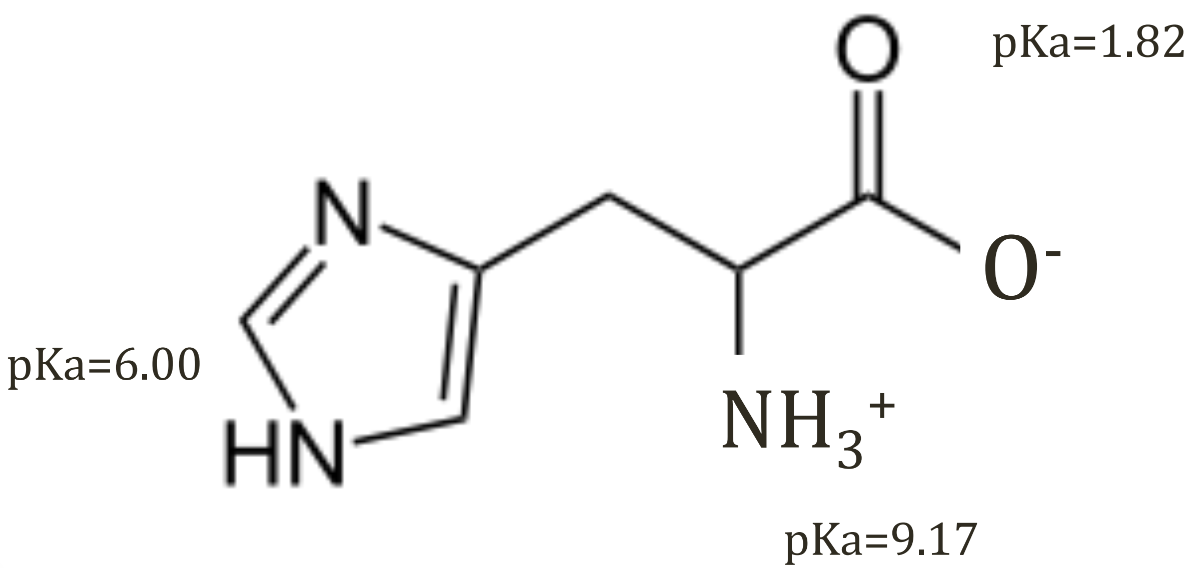
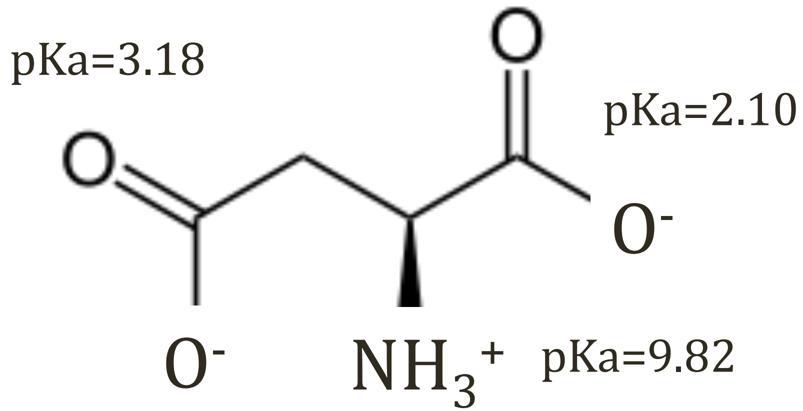
# Histidine

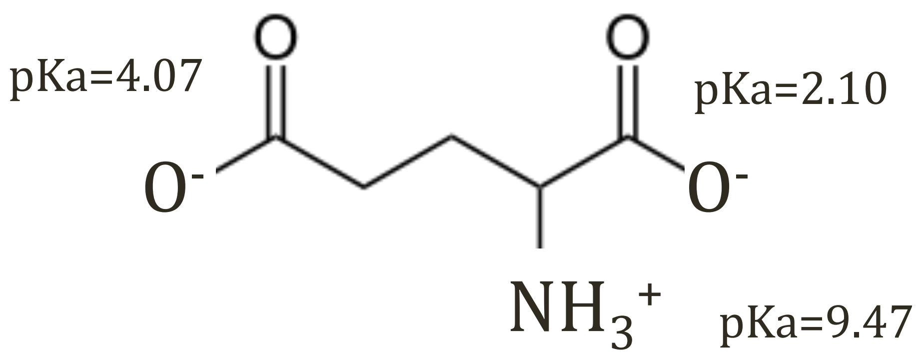
- Considered a “basic” amino acid
- Side chain pKa close to plasma pH



# Acidic Amino Acids

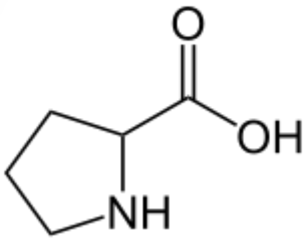



Aspartate

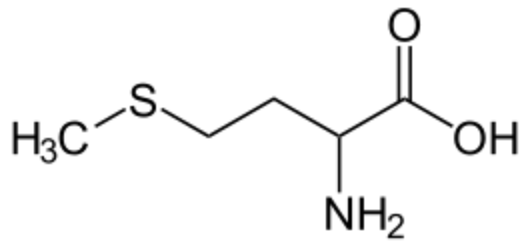



Glutamate

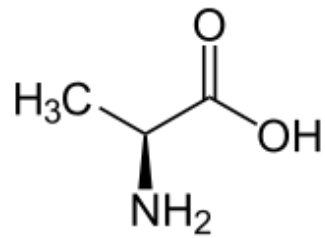
# Hydrophobic Amino Acids



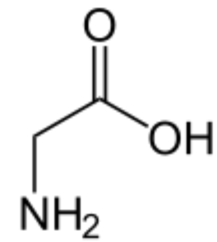

Proline



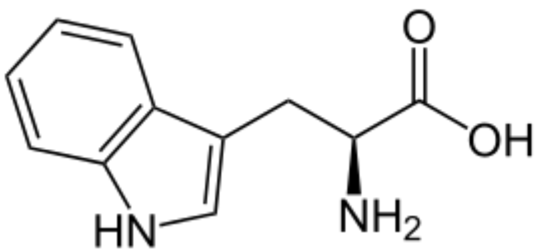

Methionine



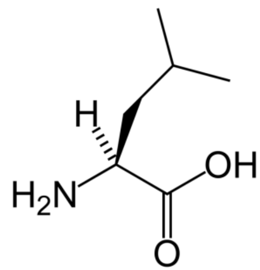

Alanine



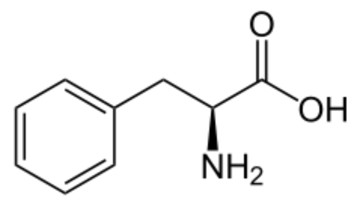

Glycine



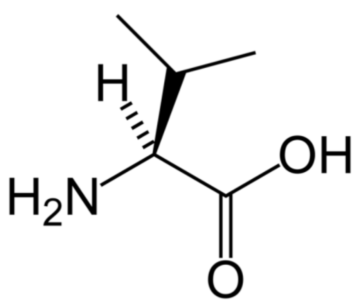

Tryptophan



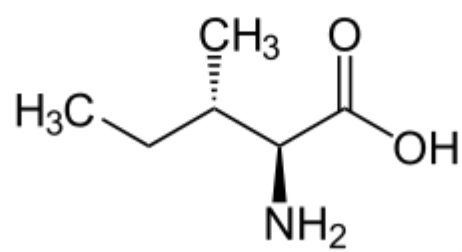

Leucine




Phenylalanine



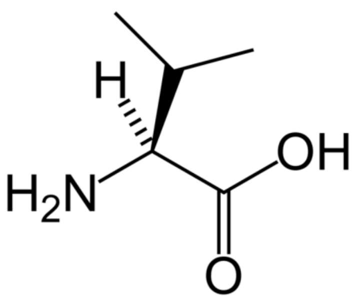

Valine



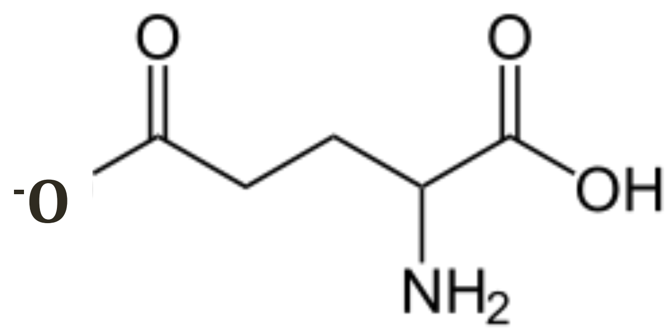

Isoleucine

# Sickle Cell Anemia

- Substitution of polar **glutamate** for nonpolar **valine** in hemoglobin protein



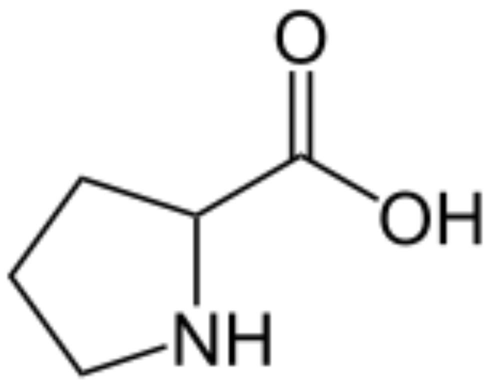

Valine




Glutamate

# Proline

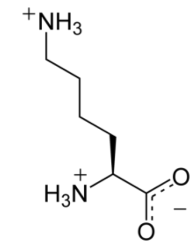
- Rigid structure (ring) formed from amino group and side chain
- Used in collagen



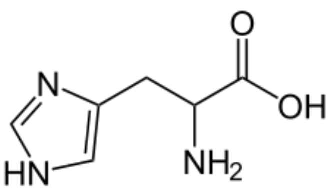

Proline

# Essential Amino Acids

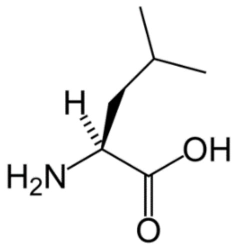
- **Nine** amino acids must be supplied by diet
- Cannot be synthesized de novo by cells



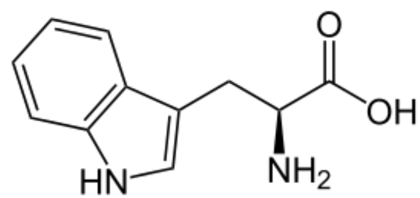
Lysine



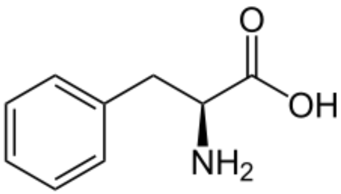
Histidine



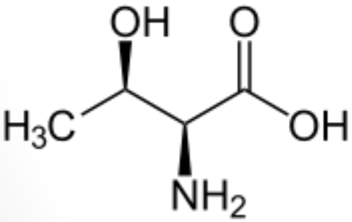
Leucine



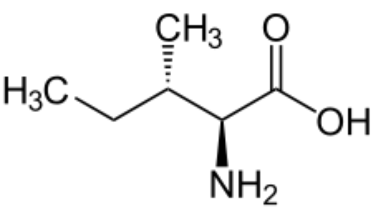
Tryptophan



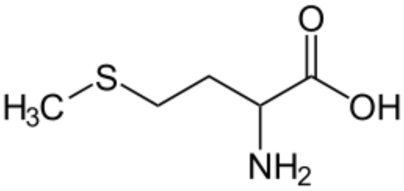
Phenylalanine



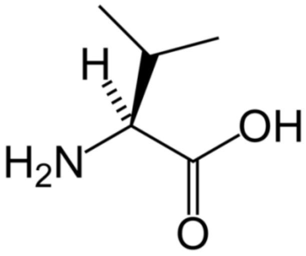
Threonine



Isoleucine



Methionine

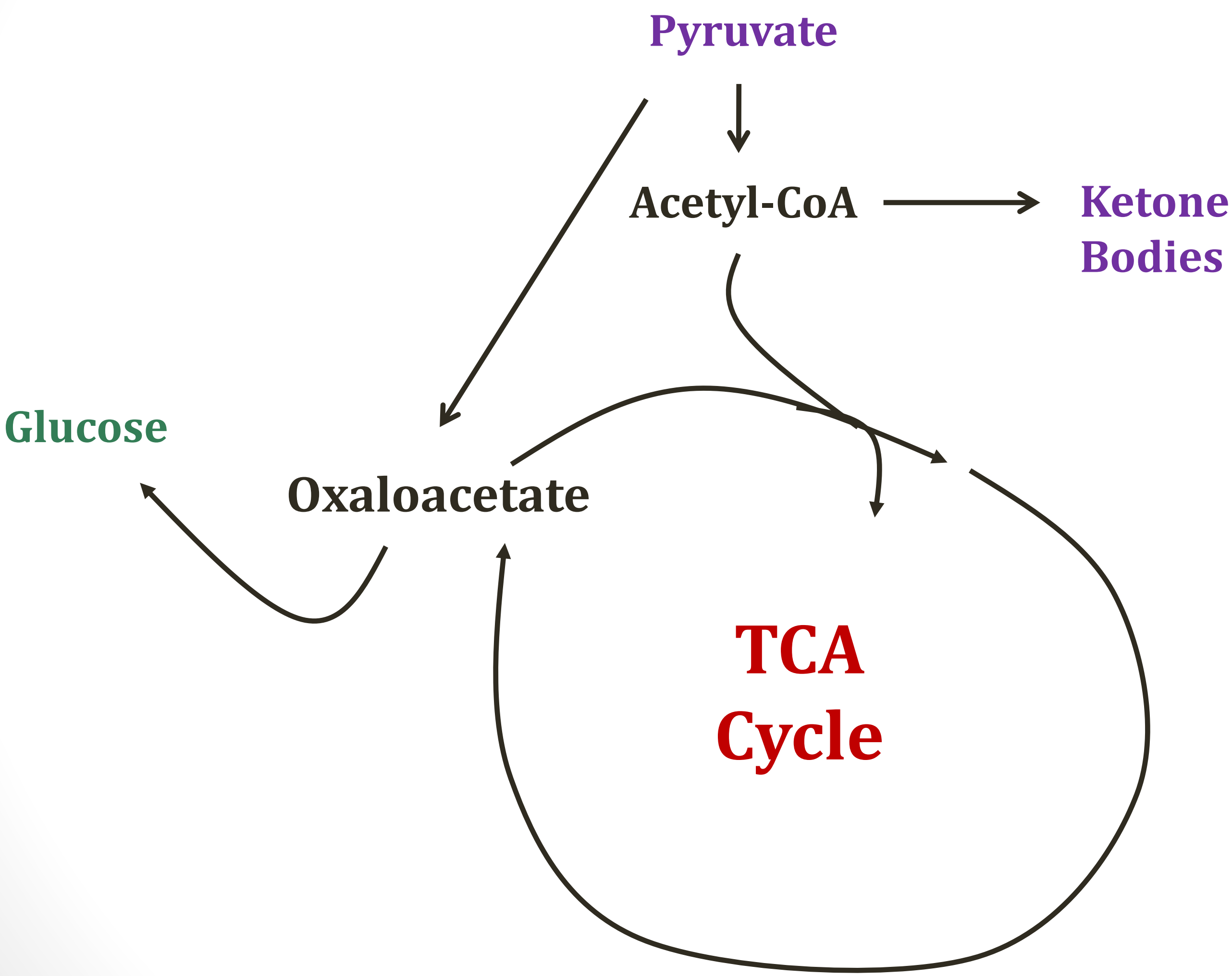


Valine

# Glucogenic vs. Ketogenic

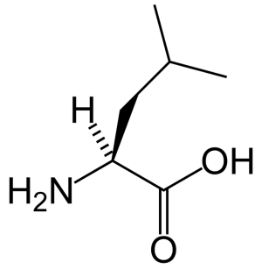
- Glucogenic amino acids:
  - Can be converted to pyruvate or TCA cycle intermediates
  - Can become glucose via gluconeogenesis
- Ketogenic amino acids
  - Convert to ketone bodies and acetyl CoA
  - Cannot become glucose
- Most amino acids are either:
  - Glucogenic
  - Glucogenic and ketogenic

# Glucogenic vs. Ketogenic

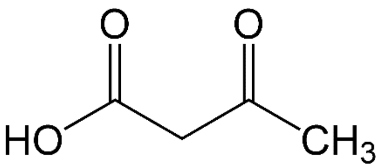


# Ketogenic Amino Acids

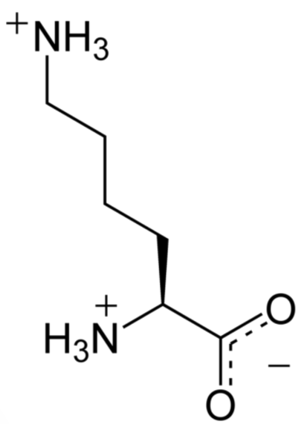
*both essential



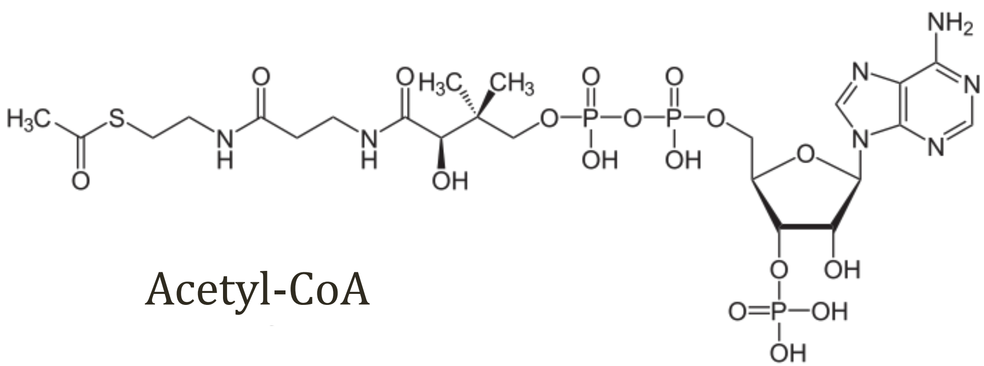
**Leucine**



Acetoacetate



**Lysine**



Acetyl-CoA

# Phenylalanine and Tyrosine

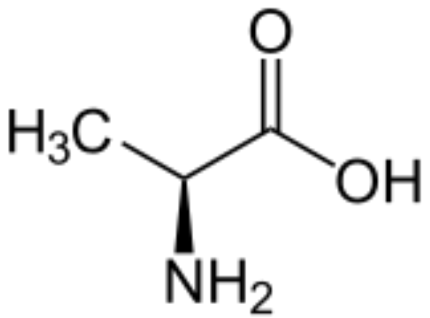
Jason Ryan, MD, MPH

# Phenylalanine and Tyrosine

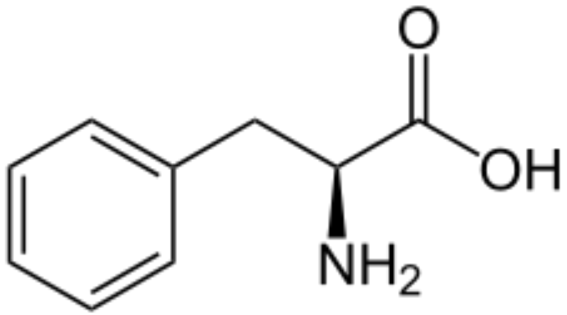
- Key amino acids for synthesis of:
  - Dopamine, Norepinephrine, Epinephrine
  - Thyroid hormone, Melanin
- Metabolism: several important vitamins/cofactors
- Three metabolic disorders:
  - Phenylketonuria (PKU)
  - Albinism
  - Alkaptonuria

# Phenylalanine

- Alanine with a phenyl group added



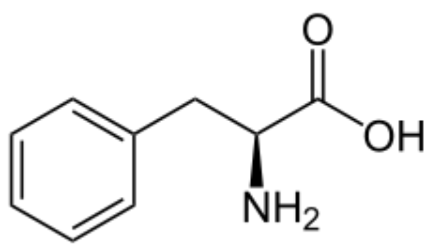

Alanine




Phenylalanine

# Phenylalanine

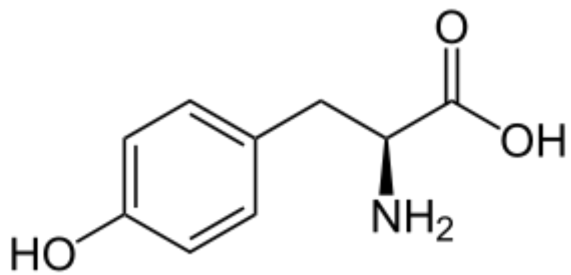
- Converted to tyrosine (non-essential amino acid)




Phenylalanine

**Phenylalanine Hydroxylase**

Tetrahydrobiopterin

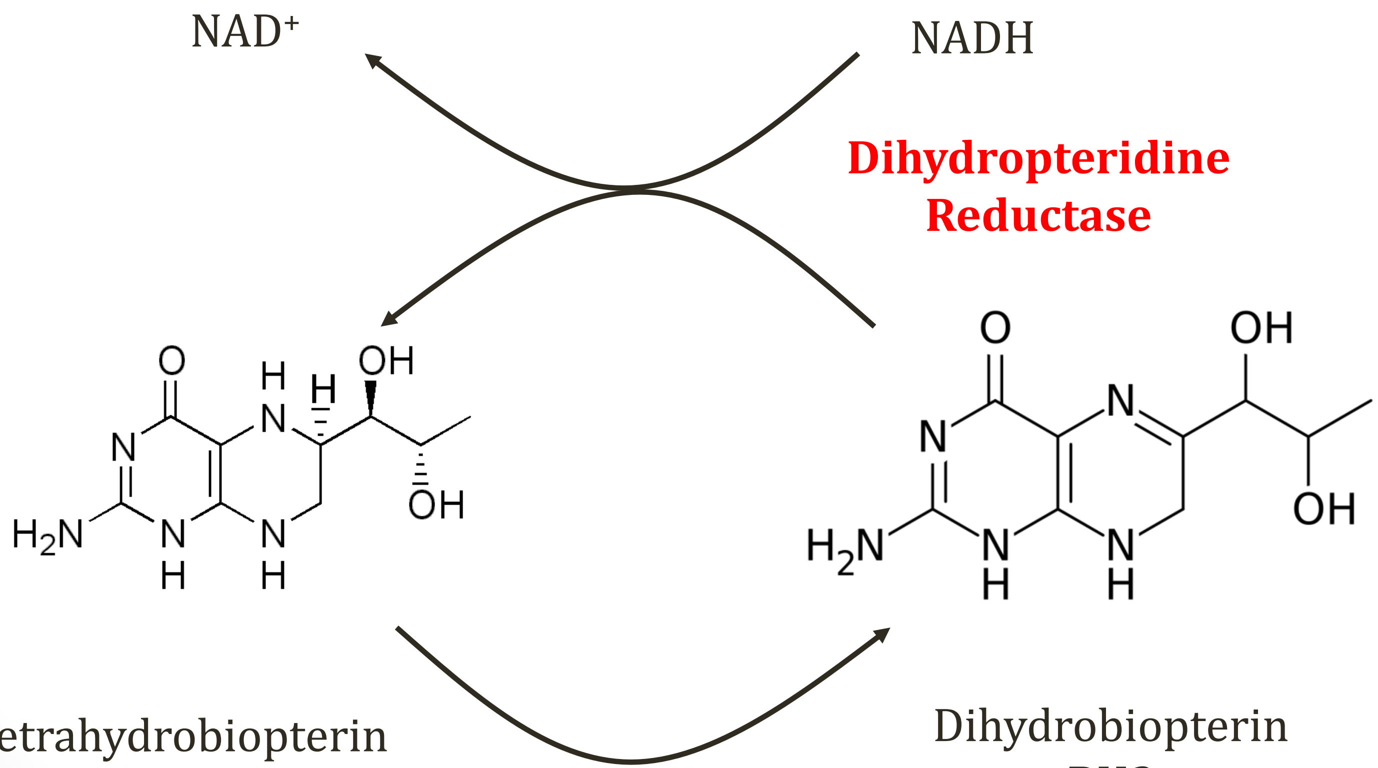



Tyrosine

# Tetrahydrobiopterin

BH4

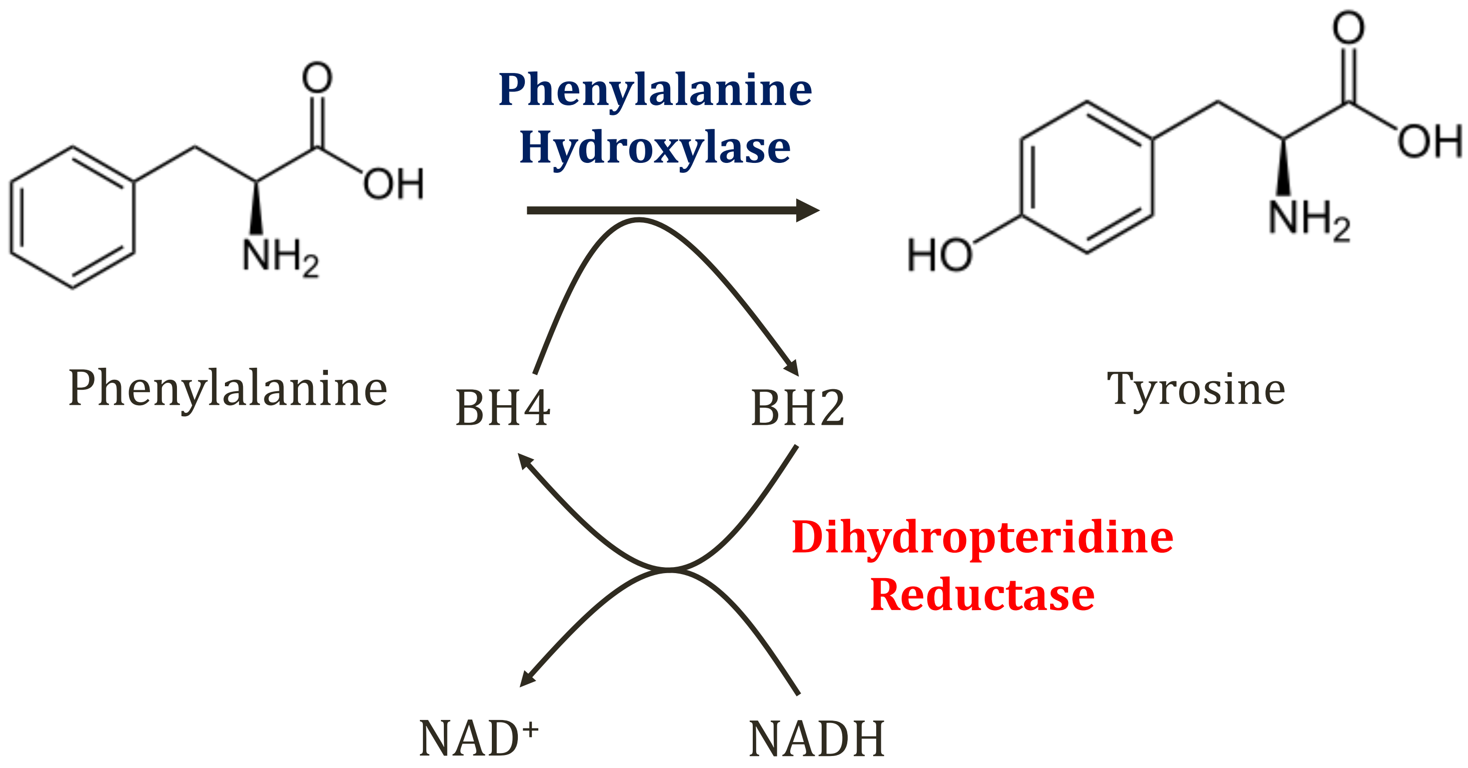
- Cofactor for phenylalanine metabolism




Tetrahydrobiopterin
BH4

Dihydrobiopterin
BH2

# Phenylalanine



# Phenylketonuria

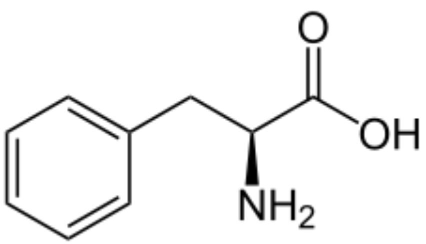
## PKU

- Deficiency of phenylalanine hydroxylase activity
  - Defective enzyme (classic PKU)
  - Defective/deficient BH4 cofactor
- Most common inborn error of metabolism
- Accumulation of phenylalanine
- Deficiency of tyrosine (sometimes low normal)

# Phenylketonuria

## PKU

- Metabolites of phenylalanine → toxicity




Phenylalanine → Phenylpyruvate → Phenyllactate / Phenylacetate

**Phenylketones**

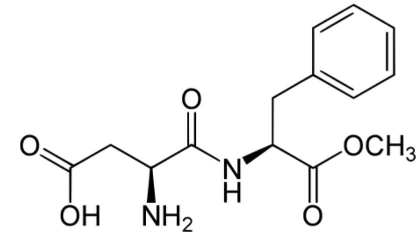
# Phenylketonuria

## Signs and Symptoms

- Musty smell in urine from phenylalanine metabolites
- CNS Symptoms
  - Mental retardation
  - Seizures
  - Tremor
- Pale skin, fair hair, blue eyes
  - Lack of tyrosine conversion to melanin

# Phenylketonuria

## Treatment




Aspartame
(aspartate + phenylalanine)

- Dietary modification
  - **Restriction of phenylalanine**
  - Found in most proteins (essential amino acid)
  - Synthetic amino acids mixtures use for food
  - Phenylalanine level monitored
  - **No aspartame** (Equal/NutraSweet)
  - **Tyrosine becomes essential**

# Phenylketonuria

- Maternal PKU
  - Occurs in **women with PKU** who consume phenylalanine
  - High levels of phenylalanine acts as a **teratogen**
  - Baby born with microcephaly, congenital heart defects

Øyvind Holmstad/Wikipedia

# Phenylketonuria

## Screening

- Newborn measurement of phenylalanine level
- Done 2-3 days after birth
  - Maternal enzymes may normalize levels at birth

Achoubey/Wikipedia

# Phenylketonuria

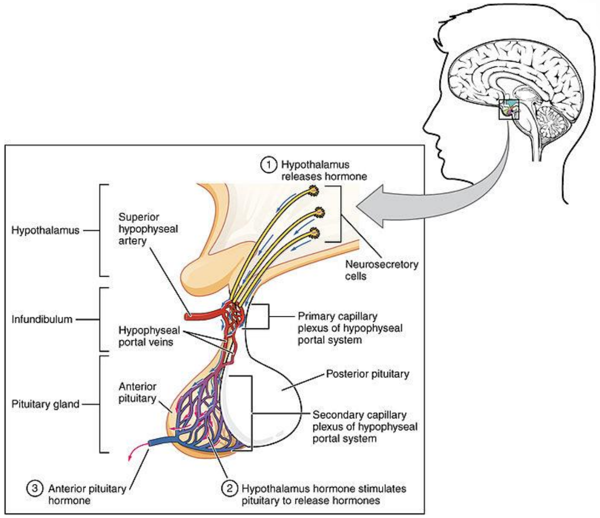
## BH4 Deficiency

- Rare (2%) cause of PKU
- Defective BH4
  - Often due to defective **dihydropteridine reductase**
  - Also impaired BH4 synthesis

~~BH4~~

# Phenylketonuria
## BH4 Deficiency

- Elevated phenylalanine
- Also decreased synthesis of:
  - Epinephrine, Norepinpehrine
  - Serotonin
  - **Dopamine (↑prolactin)**



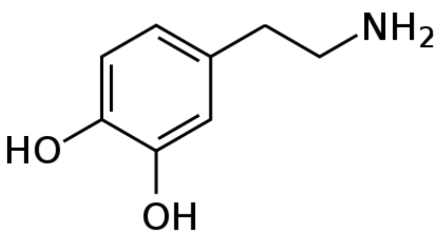

Open Stax College/Wikipedia

# Phenylketonuria

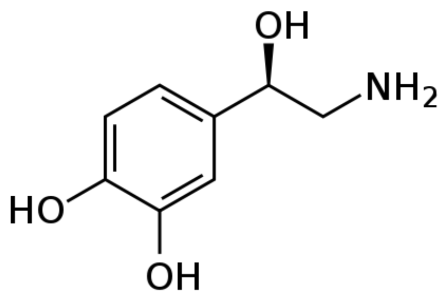
## BH4 Deficiency

- Treatment:
  - Dietary restriction of phenylalanine
  - Tyrosine supplementation (now essential)
  - Supplementation of BH4
  - **L-dopa, carbidopa** → dopamine
  - **5-hydroxytryptophan** → serotonin

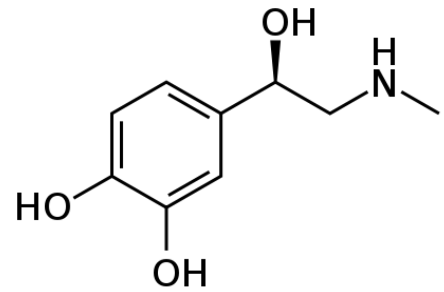
# Tyrosine Hormones



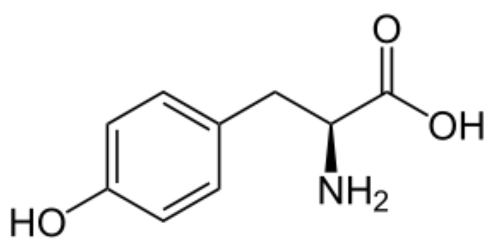

Dopamine




Norepinephrine

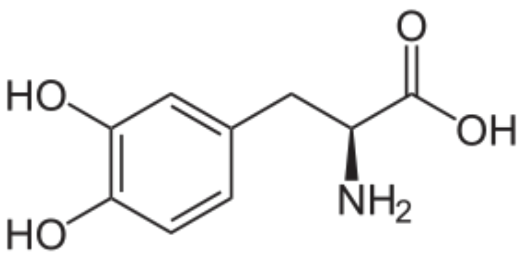


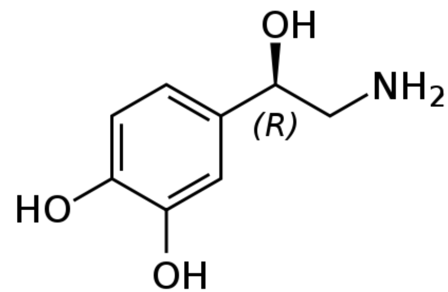

Epinephrine



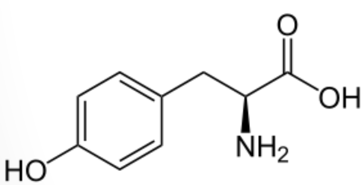

Tyrosine

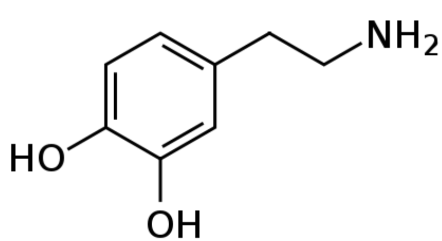
# Tyrosine Metabolism

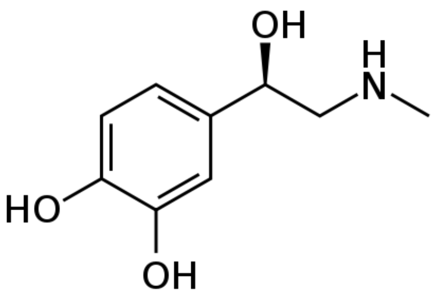




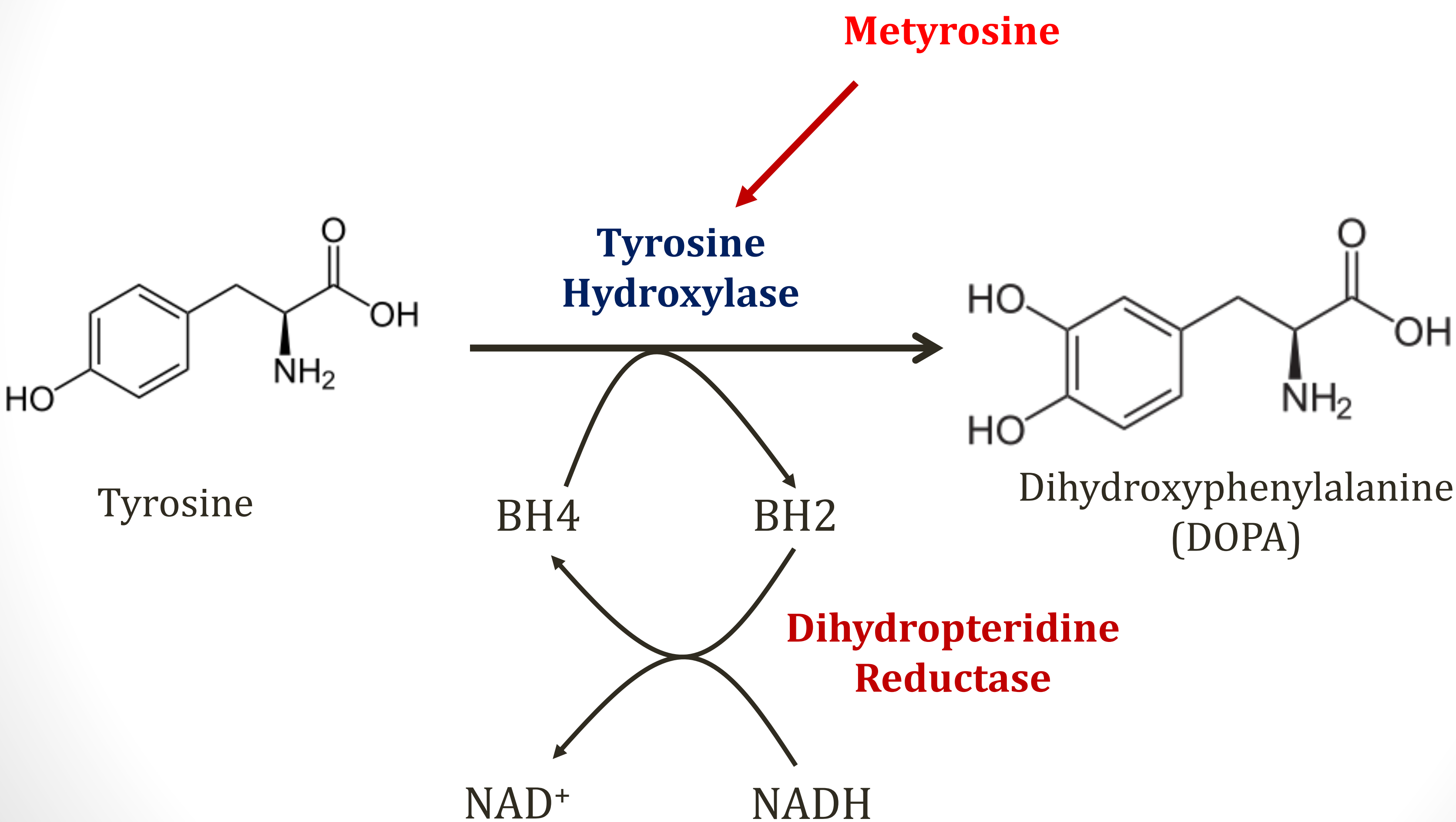
**Tyrosine → DOPA → Dopamine → Norepinephrine → Epinephrine**



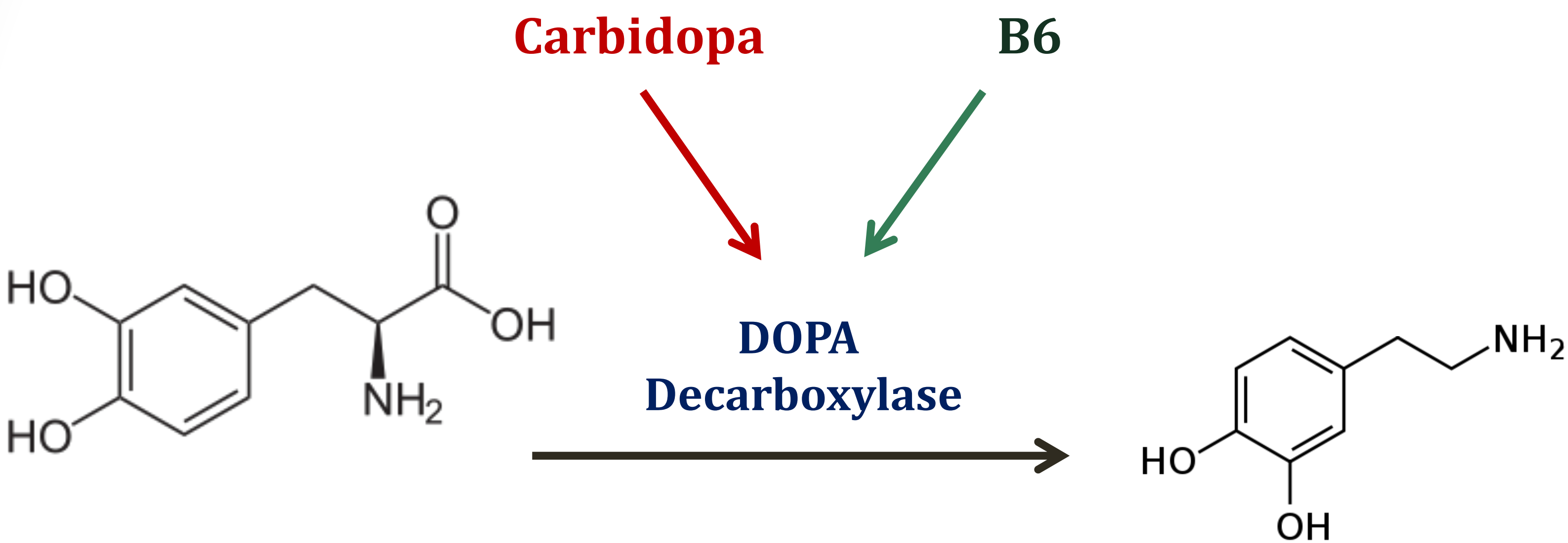




# Tyrosine Metabolism



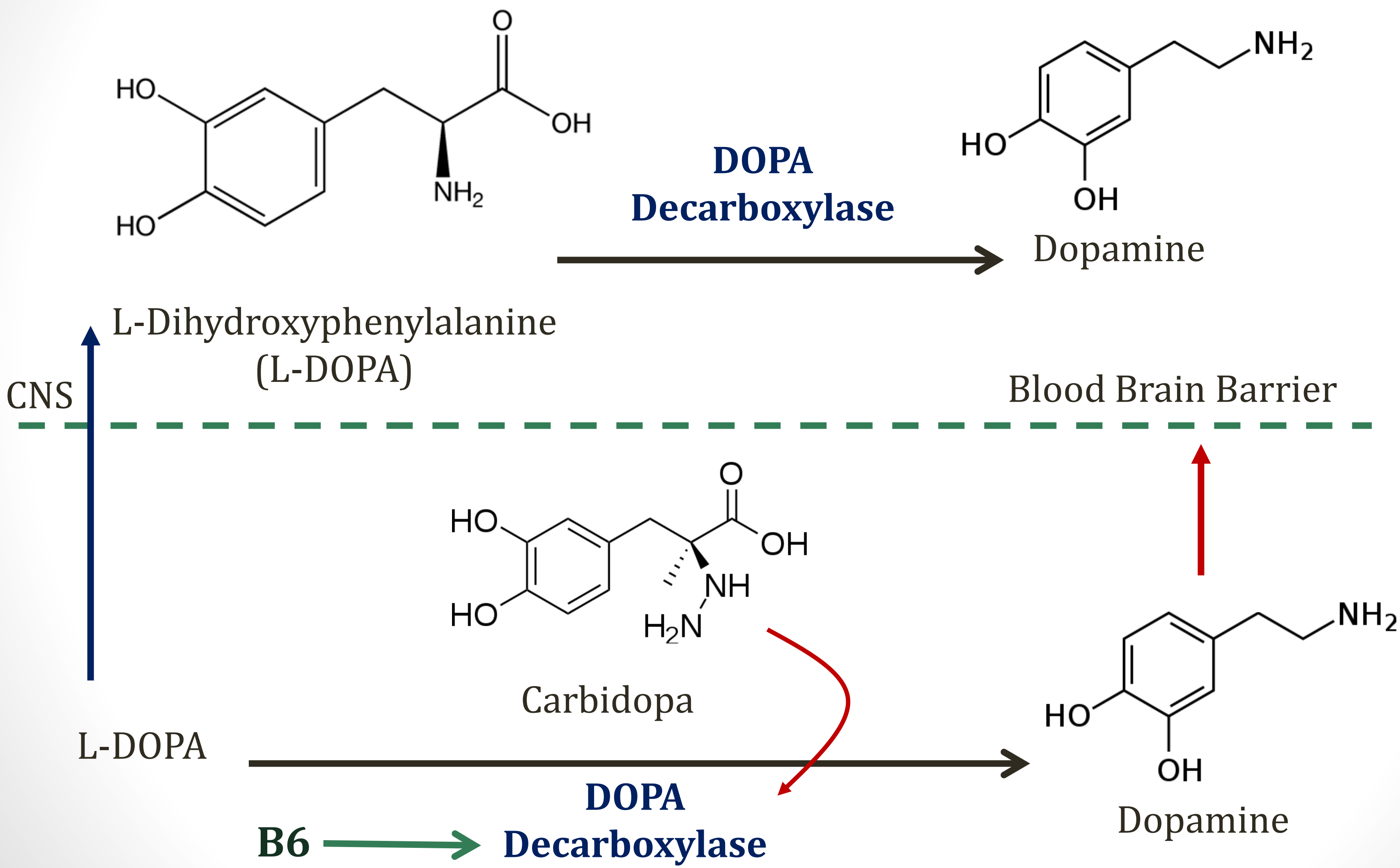
# Tyrosine Metabolism



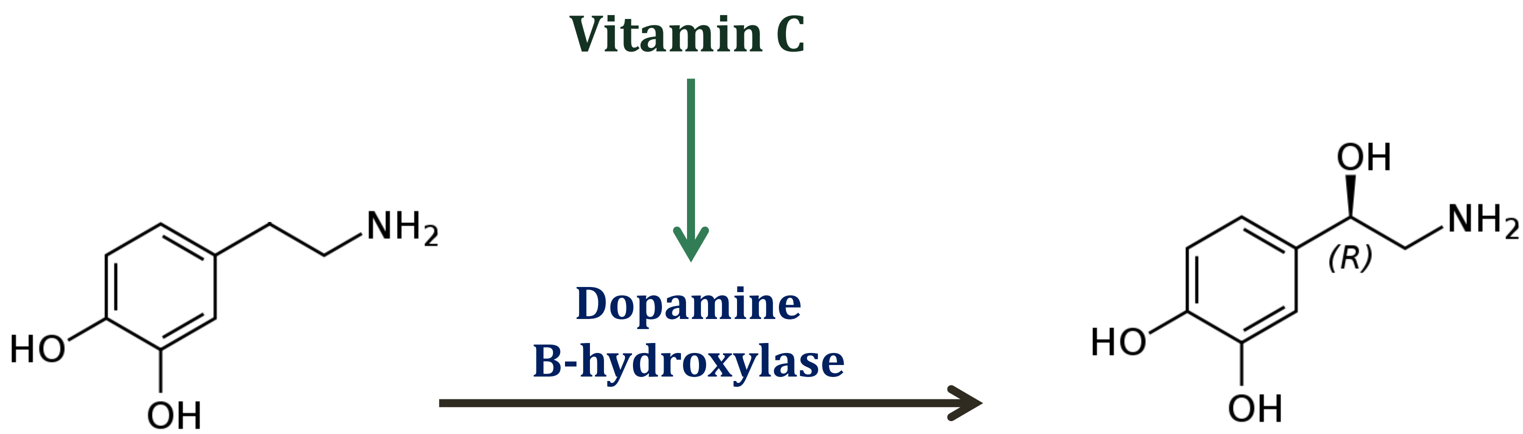

Dihydroxyphenylalanine
(DOPA)

Dopamine

# Levodopa/Carbidopa



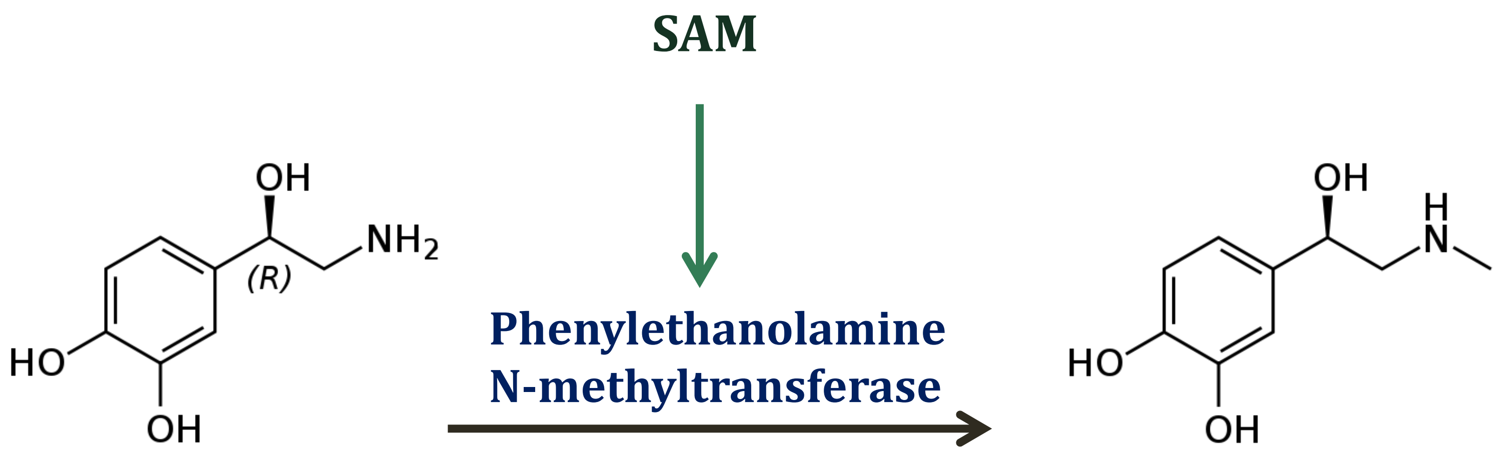
# Tyrosine Metabolism




Dopamine

Norepinephrine

# Tyrosine Metabolism



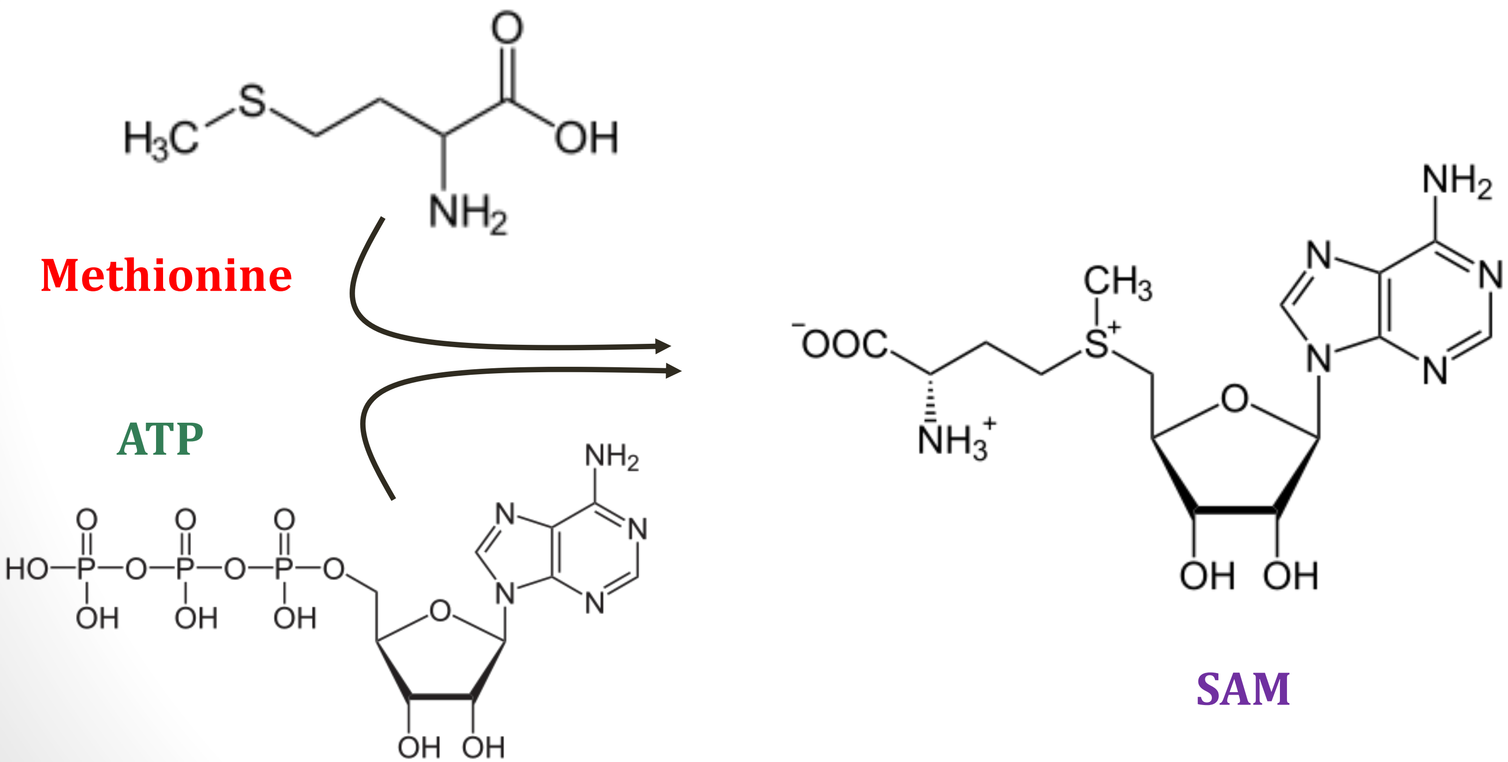

Norepinephrine

Epinephrine

# S-Adenosyl Methionine

## SAM

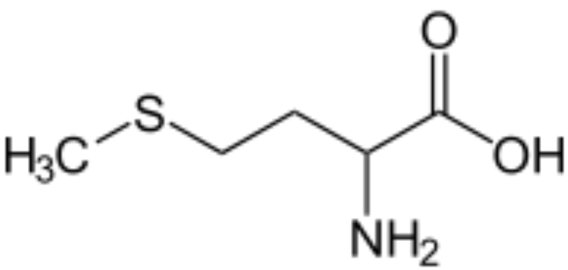
- Cofactor that donates **methyl groups**
- Synthesized from ATP and methionine



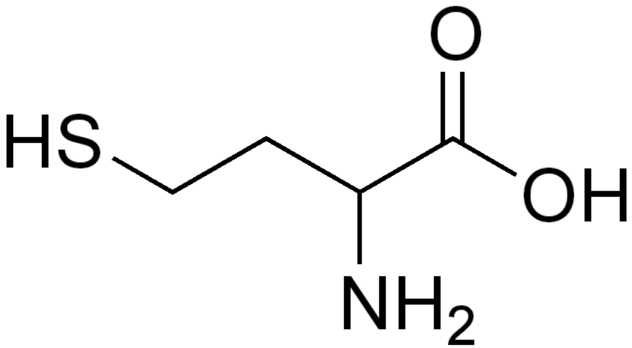
# S-Adenosyl Methionine

## SAM

- Methionine similar to homocysteine
- SAM – methyl group – adenosine = Homocysteine



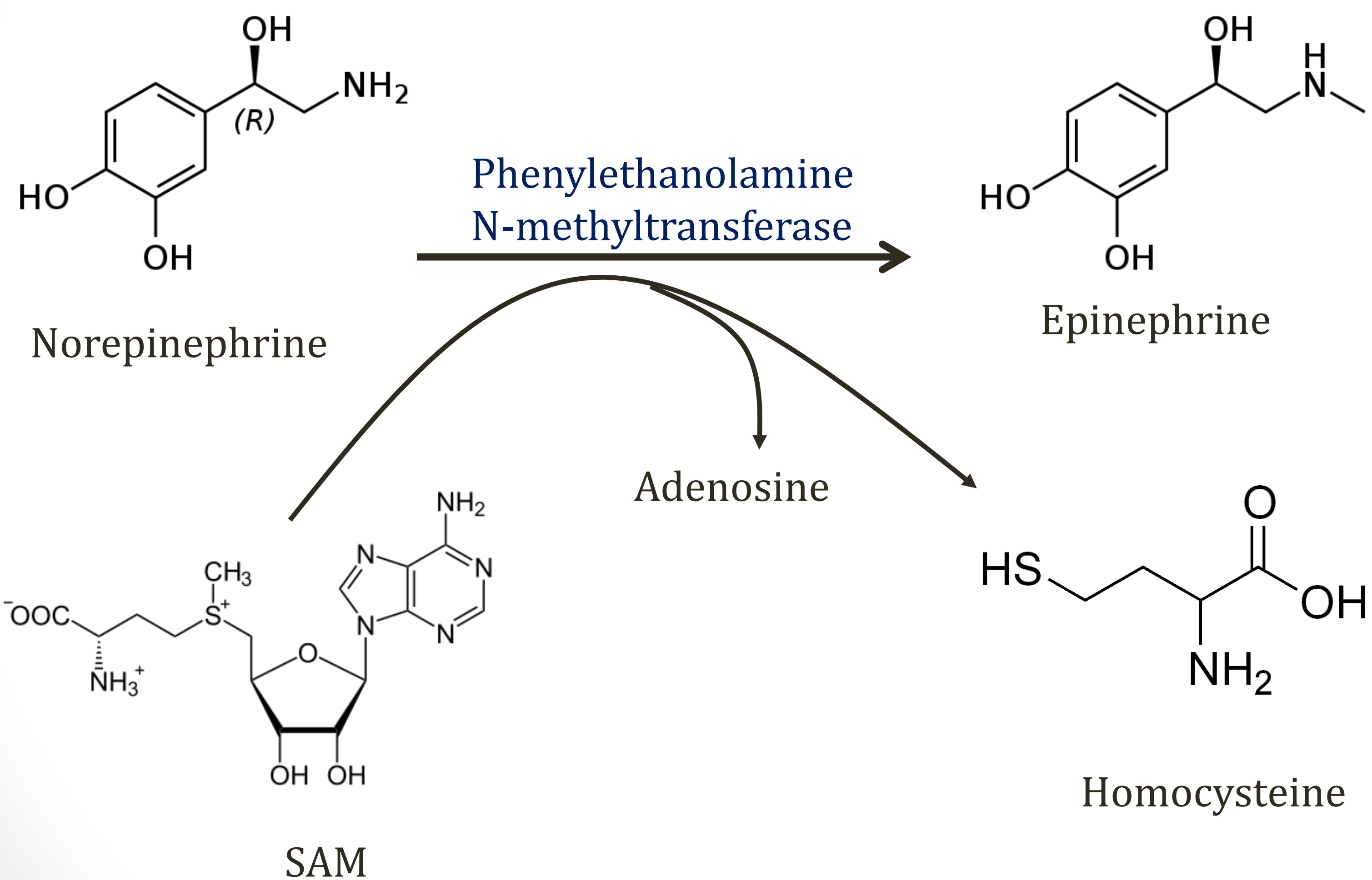

Methionine




Homocysteine

# S-Adenosyl Methionine

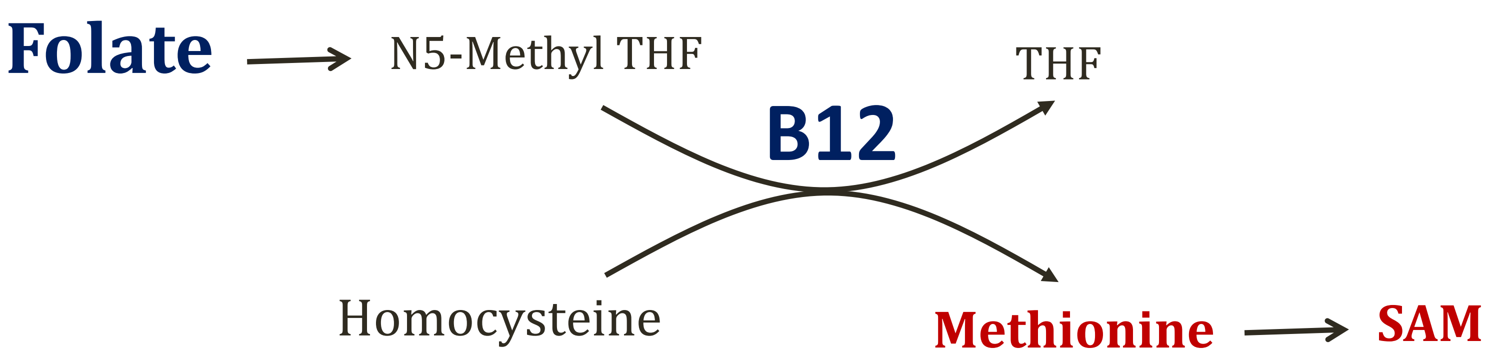
## SAM



# S-Adenosyl Methionine

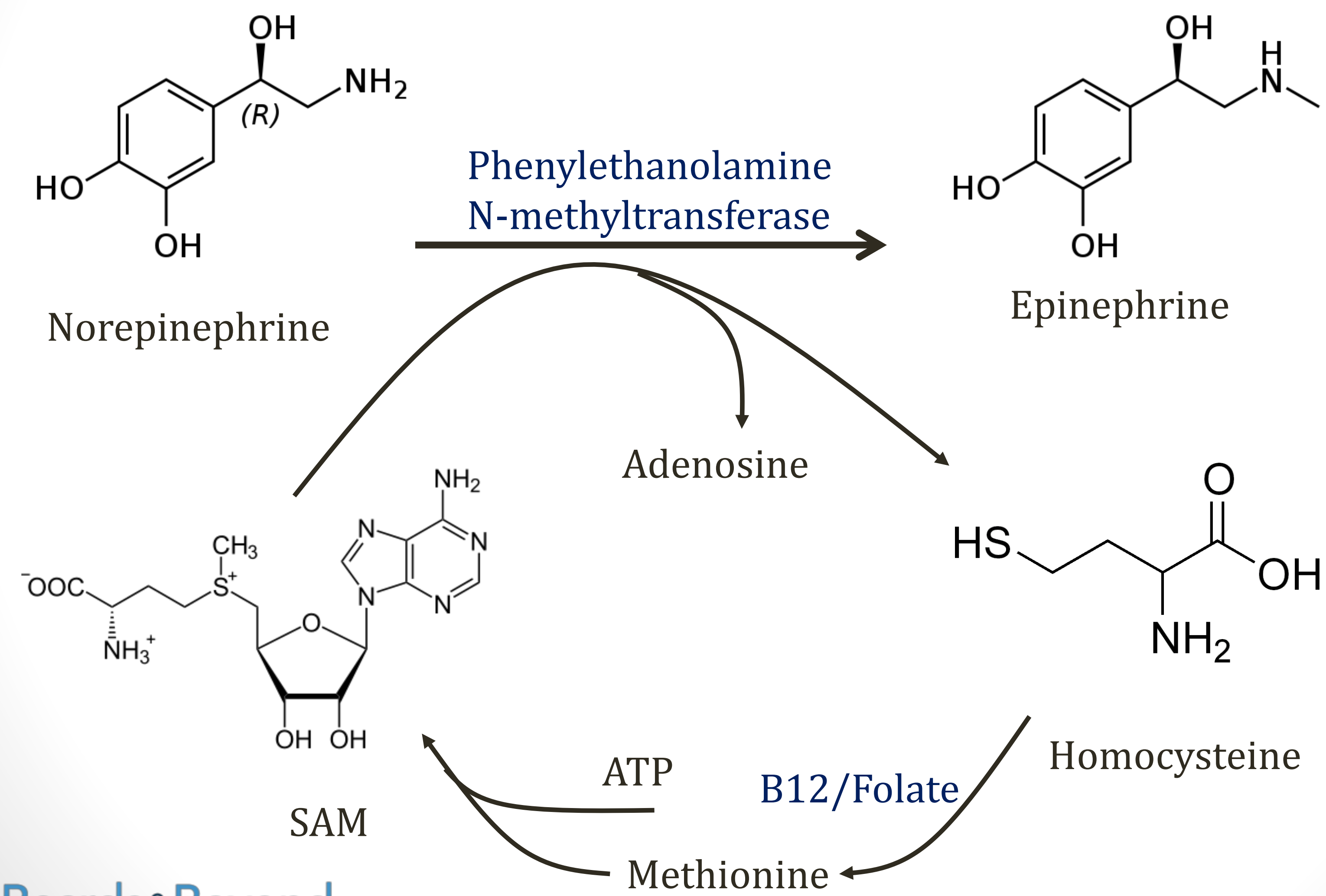
## SAM

- Need to regenerate methionine to maintain SAM
- Requires folate and vitamin B12

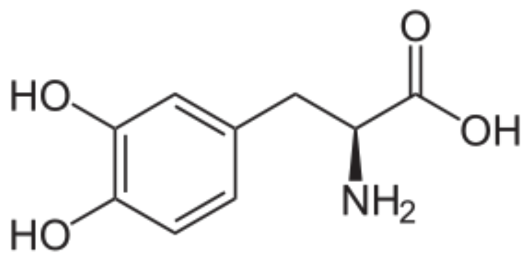


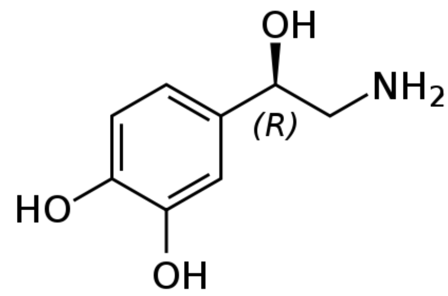
# S-Adenosyl Methionine

## SAM

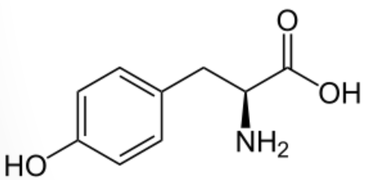


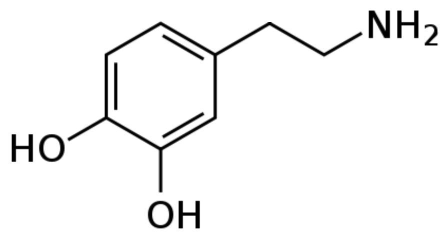
# Tyrosine Metabolism

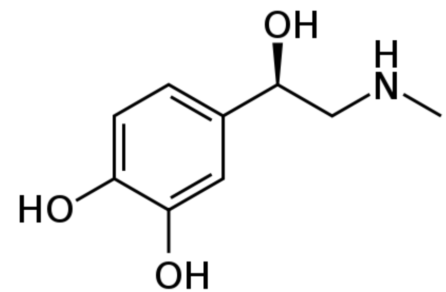


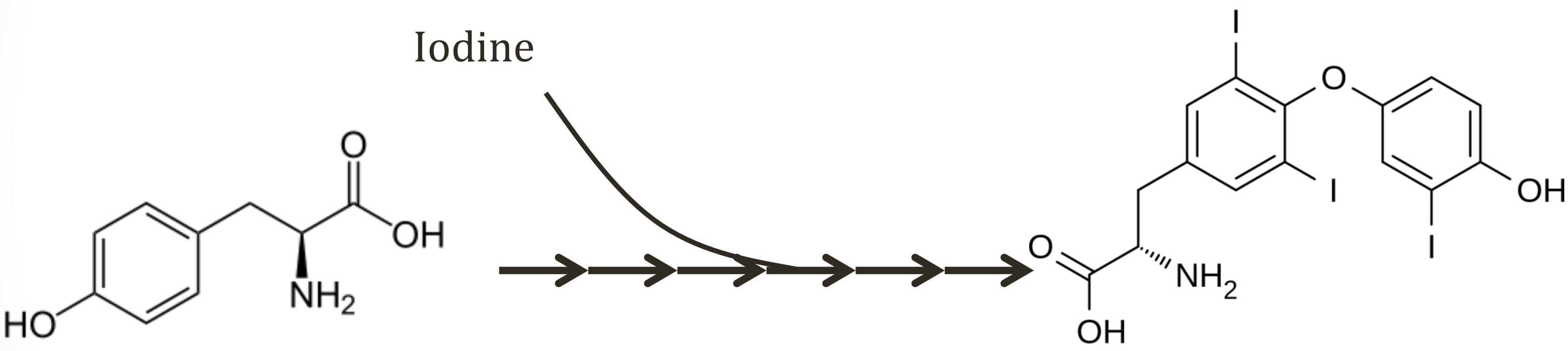

**Tyrosine** $\xrightarrow{\text{BH4}}$ **DOPA** $\xrightarrow{\text{B6}}$ **Dopamine** $\xrightarrow{\text{VitC}}$ **Norepinephrine** $\xrightarrow{\text{SAM}}$ **Epinephrine**



# Thyroxine (T4)



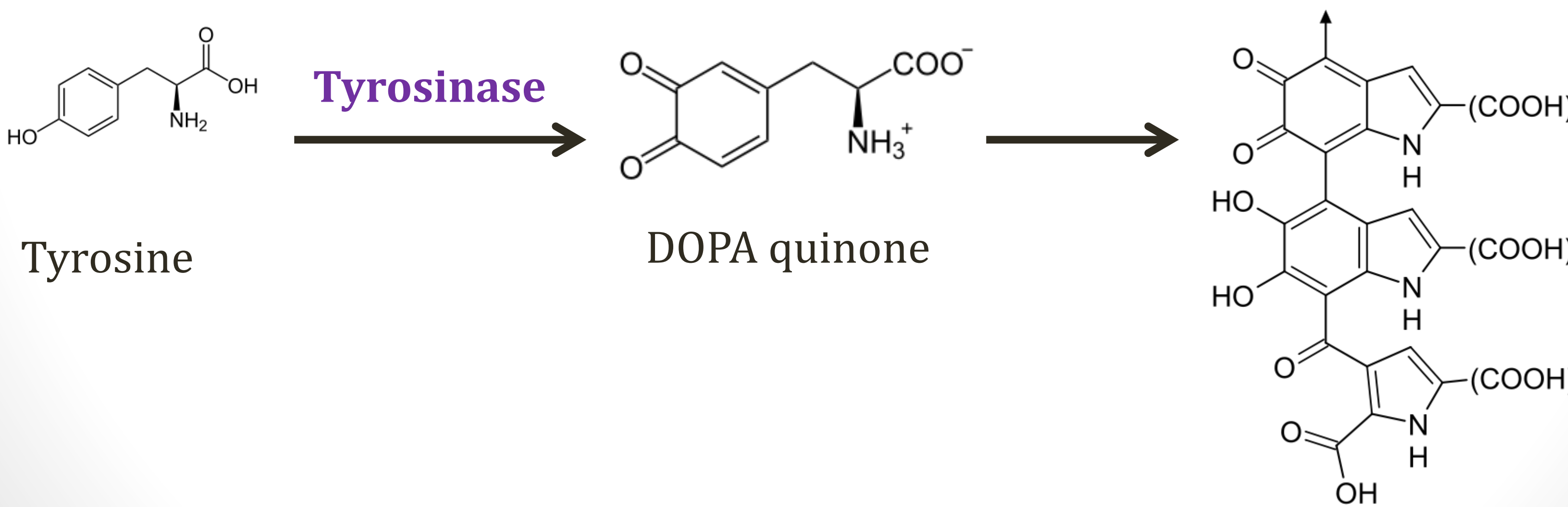

Tyrosine

Thyroxine (T4)

# Melanin

- Pigment in skin, hair, eyes
- Synthesized by melanocytes
- Polymer of repeating units made from tyrosine

Melanin




Tyrosine

DOPA quinone

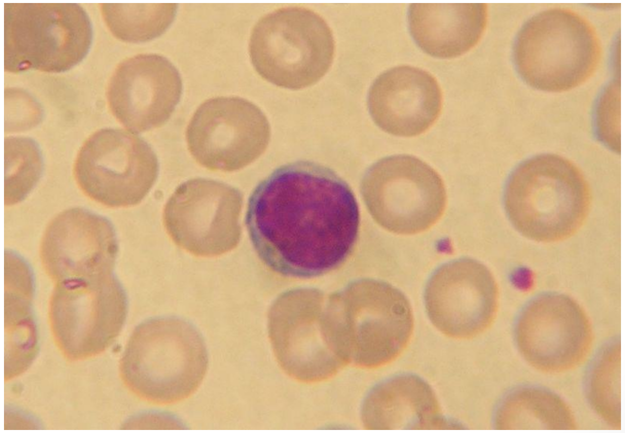
# Oculocutaneous albinism
(OCA)

- Most commonly from deficiency of:
  - Tyrosinase (OCA Type I)
  - Tyrosine transporters (OCA Type II)
- Decreased/absent melanin
- Pale skin, blond hair, blue eyes
- ↑ risk of **sunburns**
- ↑ risk of **skin cancer**

Muntuwandi/Wikipedia

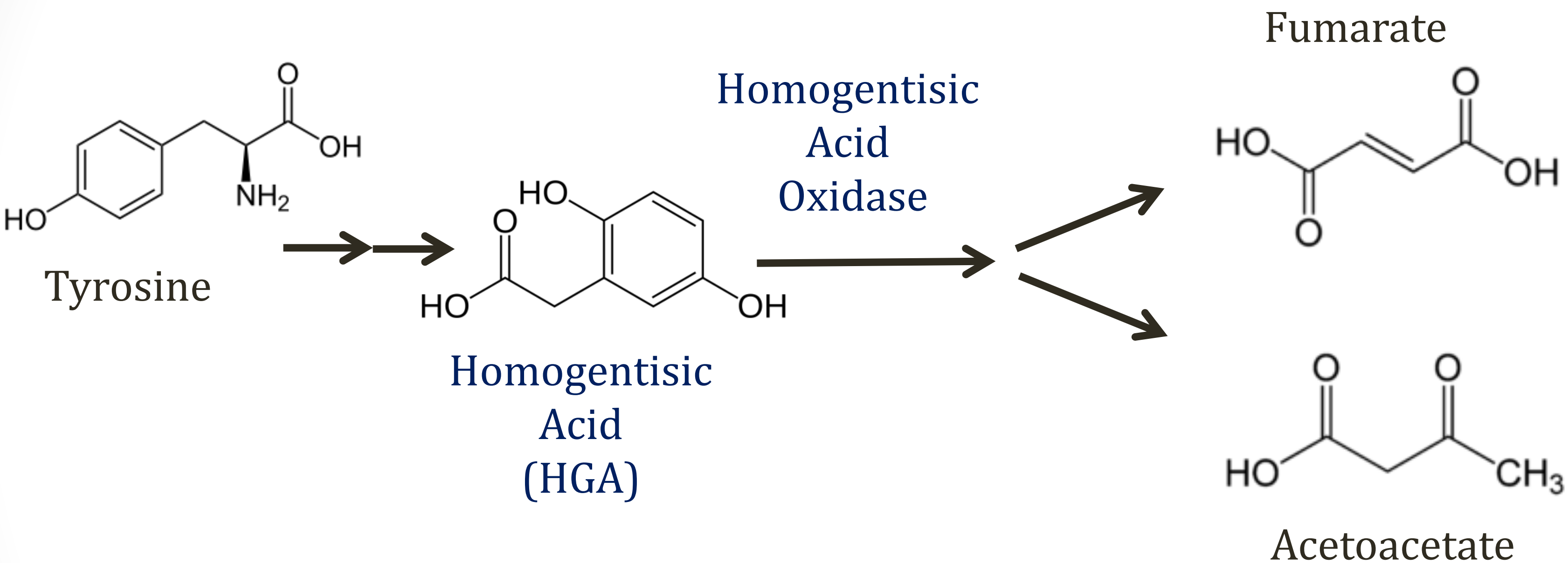
# Oculocutaneous albinism
(OCA)

- Seen in **Chediak-Higashi Syndrome**
  - Immunodeficiency
  - OCA Type II: Transporter defect
- Ocular albinism:
  - Rare variant, blue eyes only



Mgiganteus/Wikipedia

# Tyrosine Breakdown




*Tyrosine (and phenylalanine) ketogenic and glucogenic

# Alkaptonuria

## Ochronosis

- Deficiency of **homogentisic acid oxidase**
- Autosomal recessive
- ↑ homogentisic acid
- Polymerization → dark pigment
- Pigment deposited in connective tissue (ochronosis)

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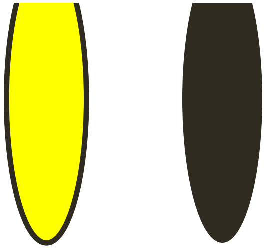
# Alkaptonuria

## Ochronosis



- Classic finding: dark urine **when left standing**
  - Fresh urine normal → polymerization
- **Arthritis** (large joints: knees, hips)
  - Severe arthritis may be crippling
- **Black pigment** in cartilage, joints
- Classic X-ray finding: calcification intervertebral discs
- Urine discoloration in infancy
- Other symptoms later in life (20-30 years)

غلامرضا باقری/Wikipedia

# Alkaptonuria

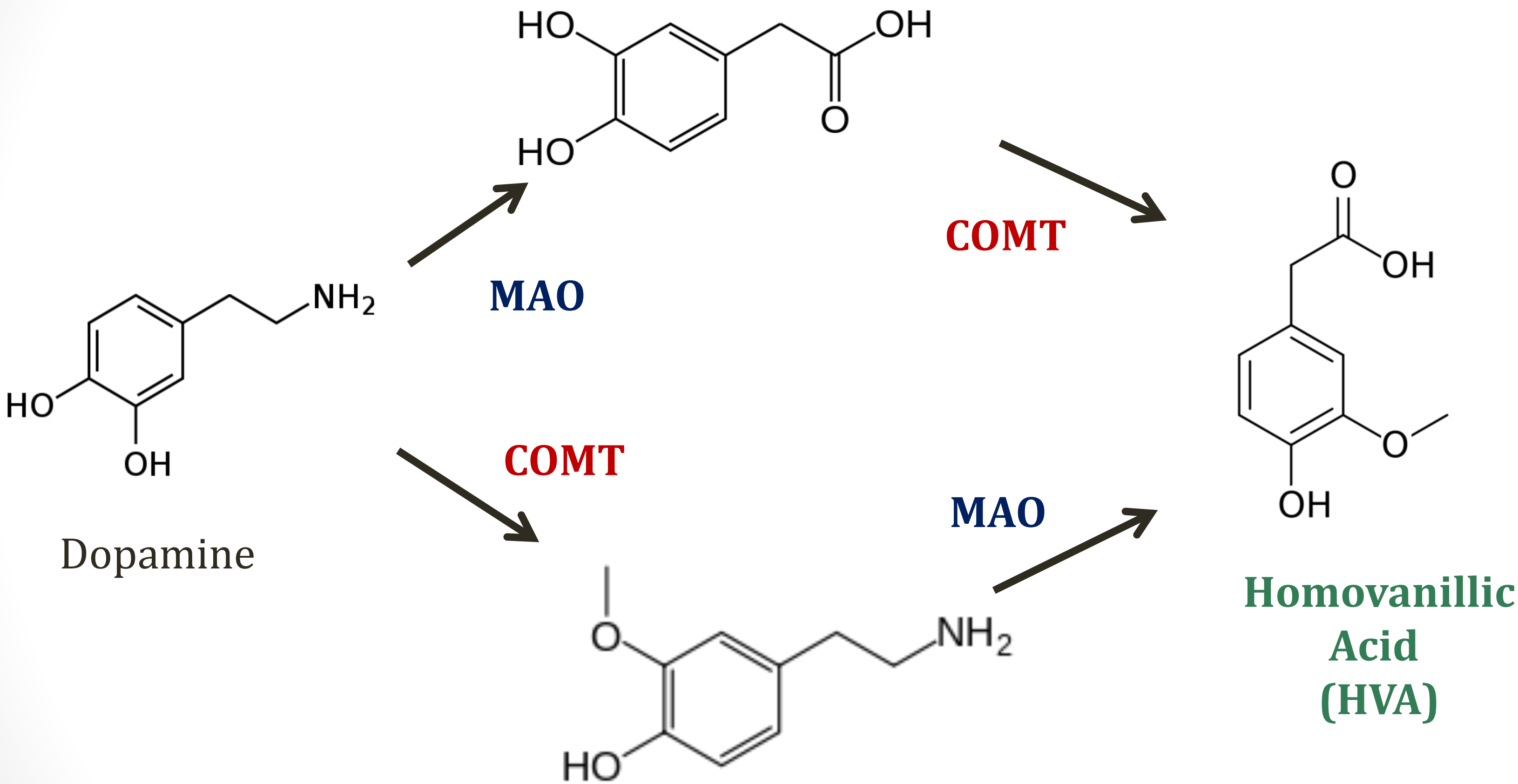
## Ochronosis

- Diagnosis
  - **Elevated HGA** in urine/plasma
- Treatment:
  - Dietary restriction (tyrosine and phenylalanine)

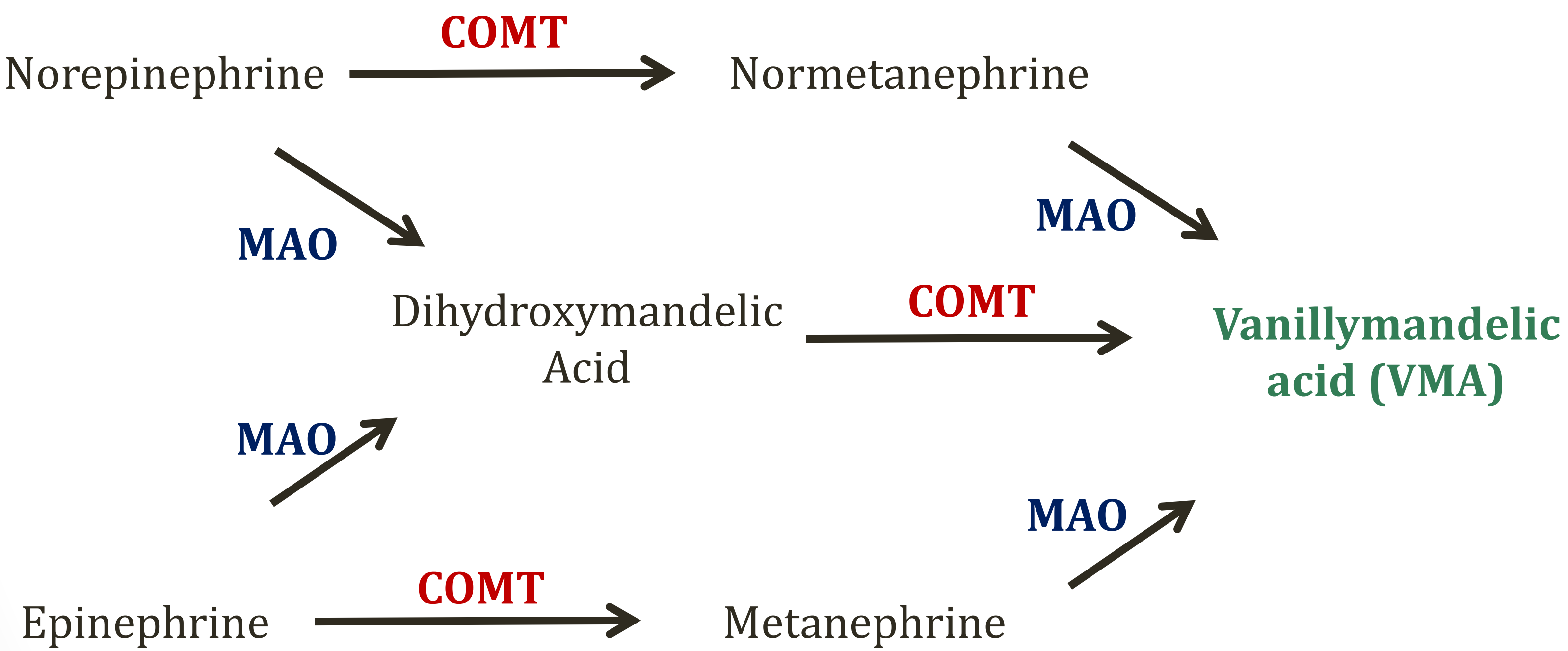
# Catecholamine Breakdown

- Monoamines: Dopamine, norepinephrine, epinephrine
- Degradation via two enzymes:
  - **Monamine oxidase (MAO)**: Amine → COOH
  - **Catechol-O-methyltransferase (COMT)**: Methyl to oxygen
- Epi, Norepi → Vanillymandelic acid (VMA)
- Dopamine → Homovanillic acid (HVA)
- HVA and VMA excreted in urine

# Tyrosine Hormones



# Tyrosine Hormones



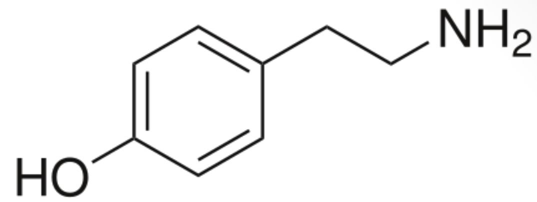
# Pheochromocytoma

- Tumor generating catecholamines
- Majority of metabolism is intratumoral
- **Metanephrines** often measured for diagnosis
  - Metanephrine and normetanephrine
  - 24hour urine collection
- Older test: 24 hour collection of VMA

# Pharmacology

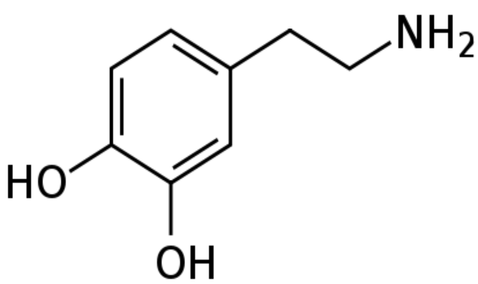
- **Parkinson's**
  - Selegiline: MAO-b inhibitor
  - Entacapone, tolacpone: COMT inhibitors
  - ↑ dopamine levels
- **Depression**
  - MAO inhibitors (Tranylcypromine, Phenelzine)
  - ↑ dopamine, NE, serotonin levels

# Tyramine

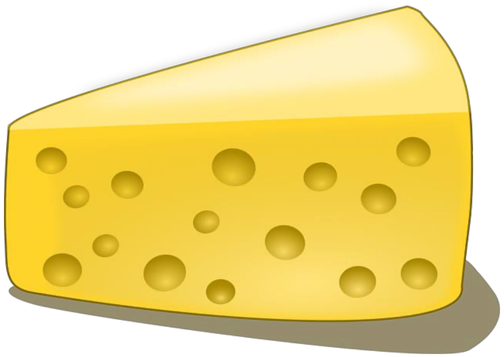



Tyramine

- Naturally occurring substance
- Sympathomimetic (causes sympathetic activation)
- Normally metabolized GI tract
- Patients on MAOi → tyramine in blood
- Hypertensive crisis
- "Cheese effect"
  - Cheese, red wine, some meats




Dopamine



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# Other Amino Acids

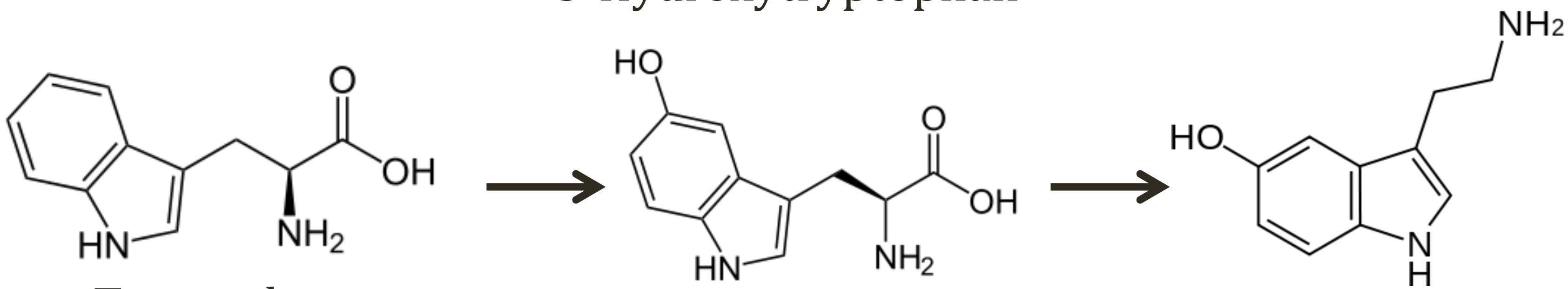
Jason Ryan, MD, MPH

# Amino Acids

- Tryptophan → Niacin, serotonin, melatonin
- Histidine → Histamine
- Glycine → Heme
- Arginine → Creatine, urea, nitric oxide
- Glutamate → GABA
- Branched chain amino acids (Maple syrup urine)
- Homocysteine (homocystinuria)
- Cysteine (cystinuria)

# Tryptophan

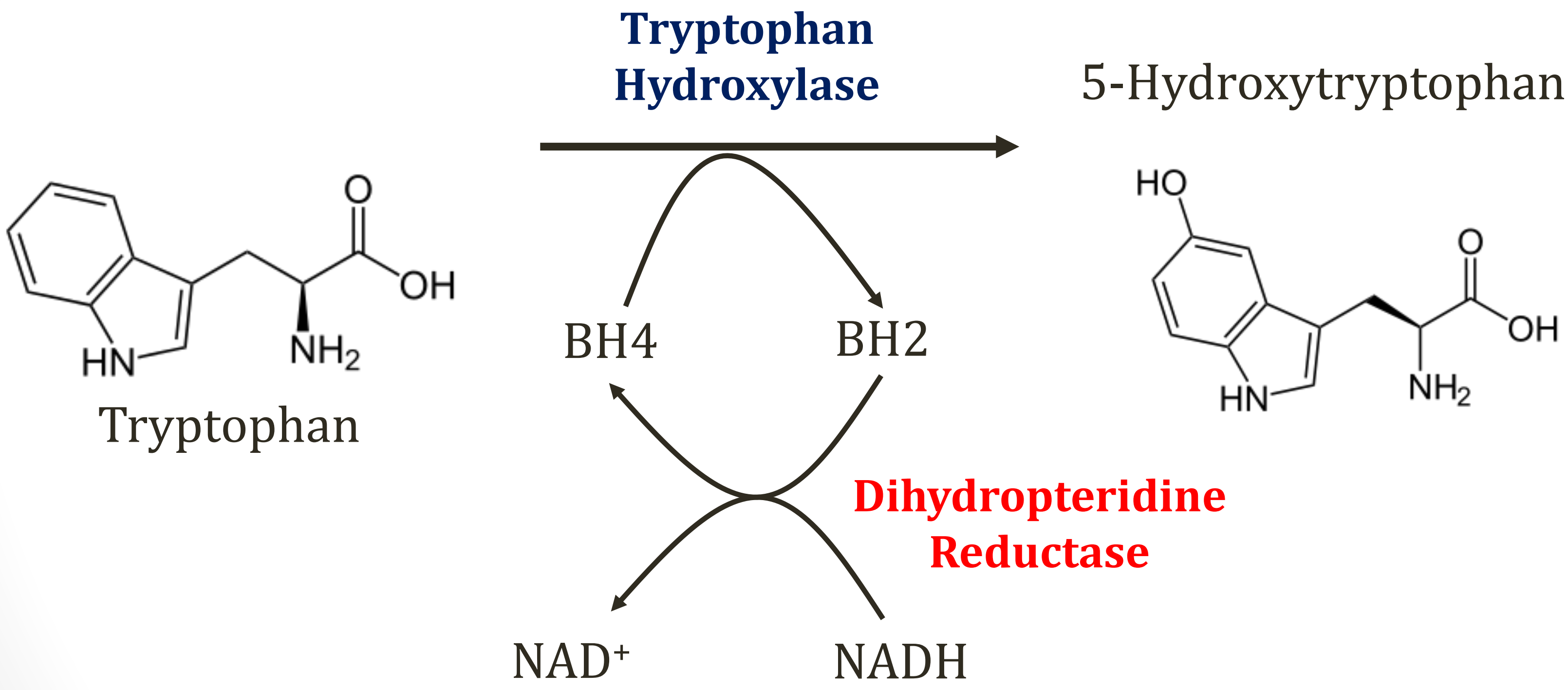
5-Hydroxytryptophan



Tryptophan

Serotonin
5-hydroxytryptamine
(5-HT)

# Tryptophan

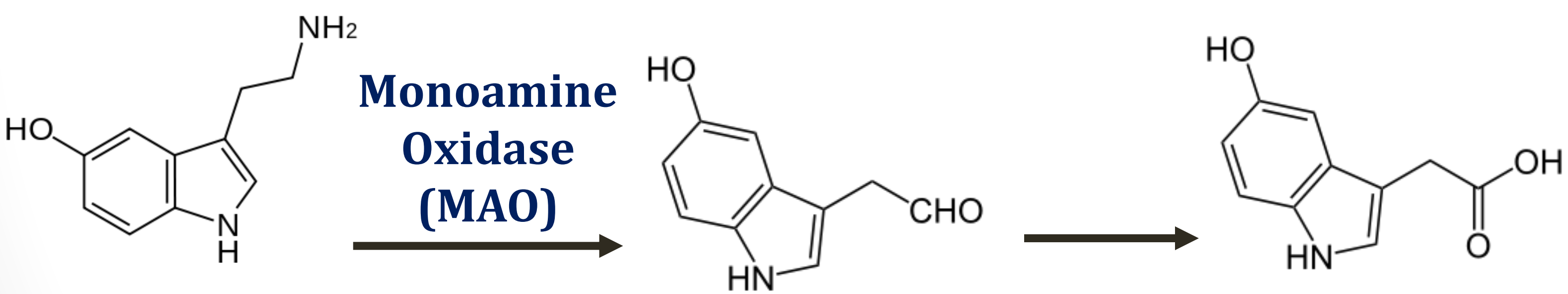

Tryptophan Hydroxylase
5-Hydroxytryptophan
HO
O
OH
HN
NH2
Tryptophan
BH4
BH2
Dihydropteridine Reductase
NAD+
NADH

# Carcinoid Syndrome

- Caused by **GI tumors** that secrete serotonin
- Altered **tryptophan metabolism**
    - Normally ~1% tryptophan → serotonin
    - Up to 70% in patients with carcinoid syndrome
    - **Tryptophan deficiency (pellagra)** reported
- Serotonin effects
    - Diarrhea (serotonin stimulates GI motility)
    - ↑ fibroblast growth and fibrogenesis → valvular lesions
    - Flushing (other mediators also)

# Serotonin Breakdown

- Metabolism via **monoamine oxidase**
  - Same enzyme: dopamine/epinephrine/norepinephrine
- **MAO inhibitors** used in depression (↑serotonin)
- ↑ **Urinary 5-HIAA** in carcinoid syndrome

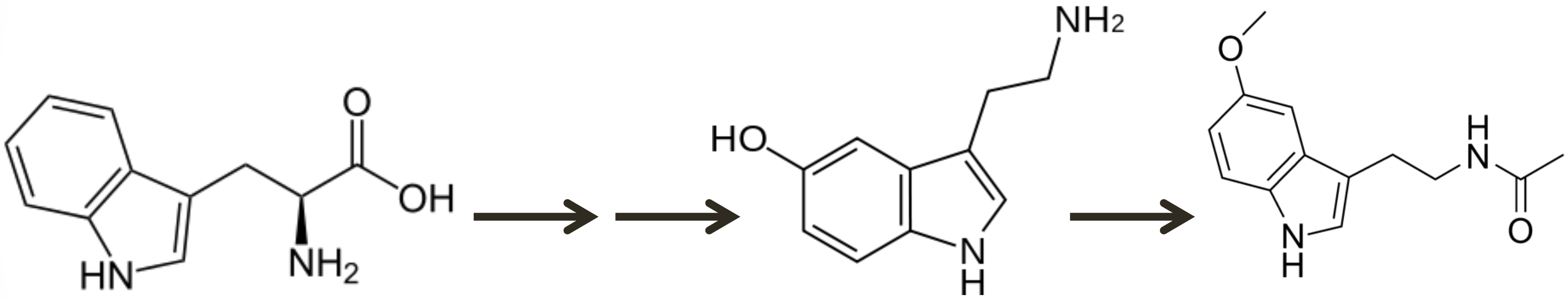



Serotonin
5-hydroxytytamine
(5-HT)

5-hydroxyindole
Acetaldehyde
(5-HT)

5-Hydroxyindoleacetic
acid
(5-HIAA)

# Tryptophan

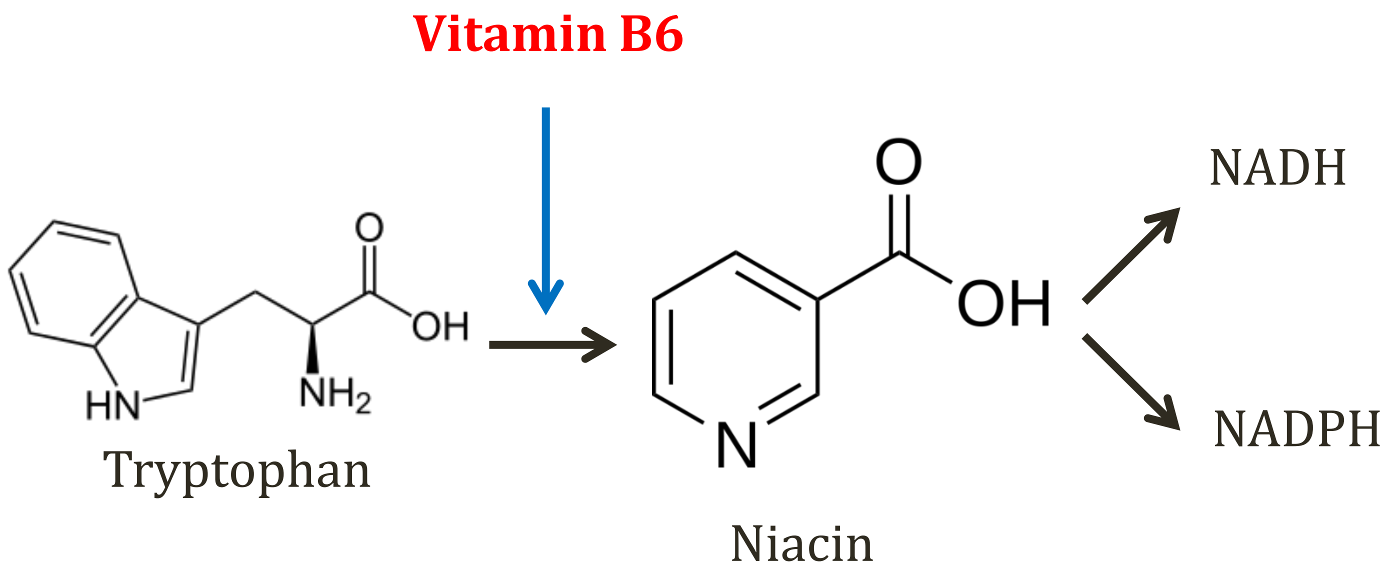



Tryptophan

Serotonin
5-hydroxytytamine
(5-HT)

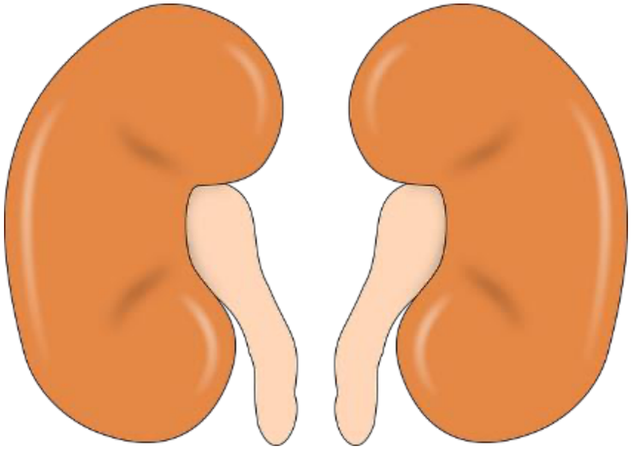
Melatonin

# Tryptophan



# Hartnup Disease

- Absence of AA transporter in **proximal tubule**
- Autosomal recessive
- Loss of **tryptophan** in urine
- Symptoms from **niacin** deficiency



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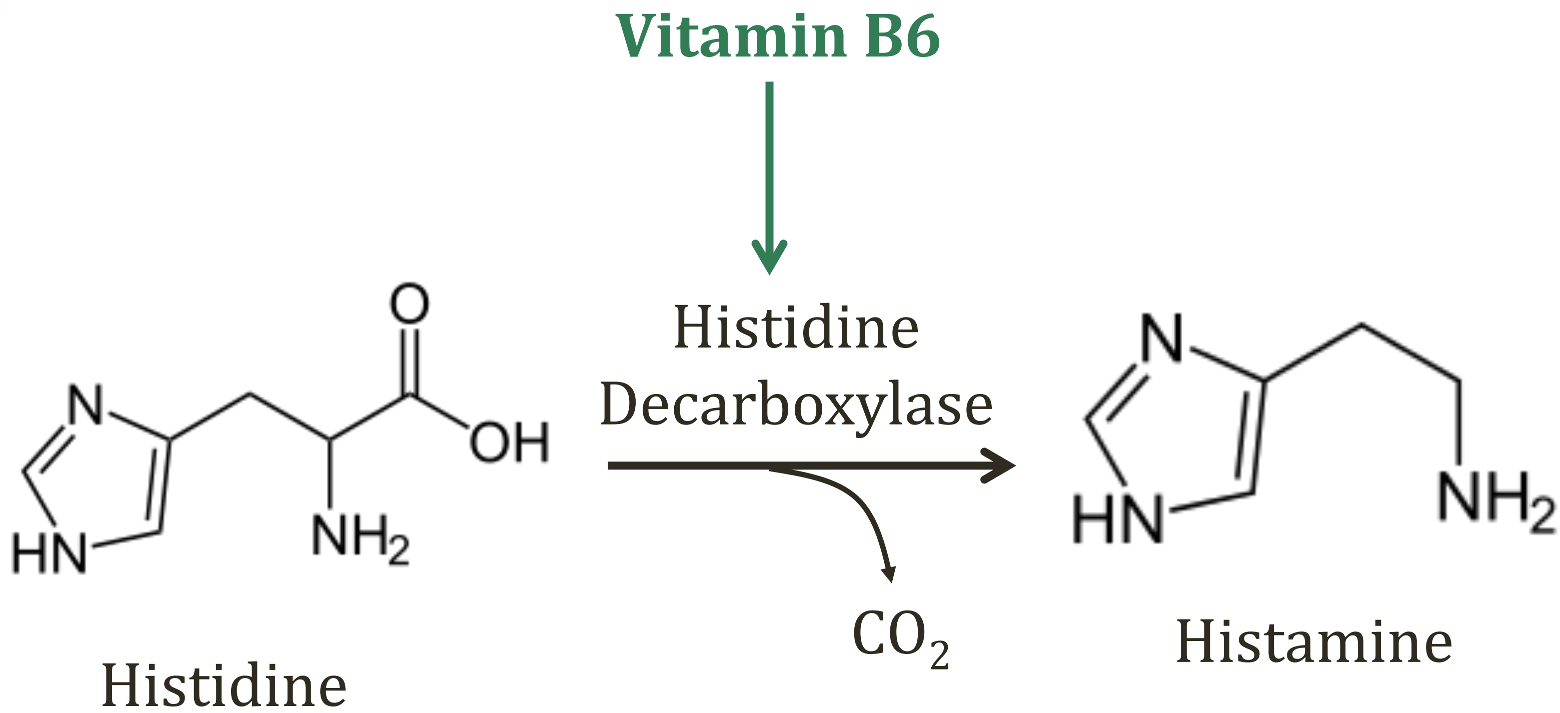
# Hartnup Disease

- Pellagra
  - Hyperpigmented rash
  - Exposed areas of skin
  - **Red tongue** (glossitis)
  - **Diarrhea** and vomiting
  - CNS: dementia, encephalopathy
  - "Dermatitis, diarrhea, dementia"
- Treatment:
  - High protein diet
  - Niacin

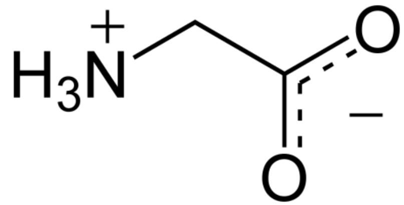
Herbert L. Fred, MD, Hendrik A. van Dijk

# Histidine



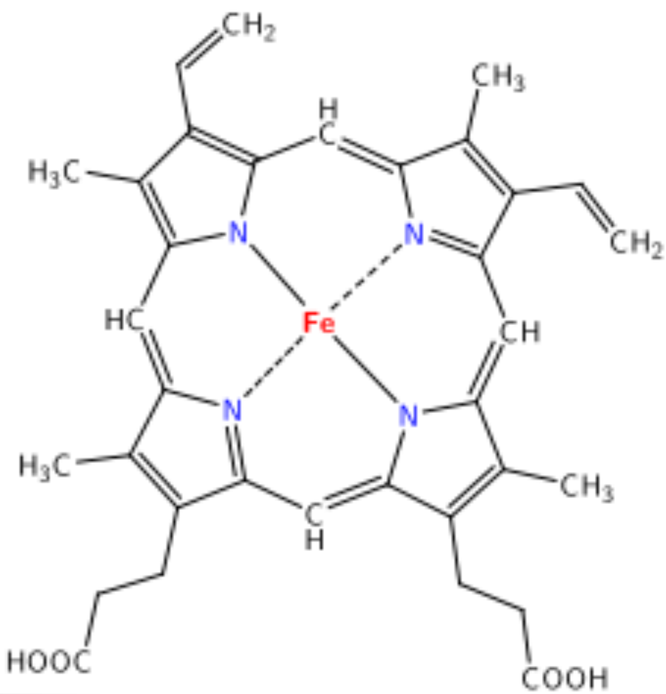
Vitamin B6
Histidine Decarboxylase
N
HN
NH2
O
OH
CO2
Histidine
Histamine

# Glycine

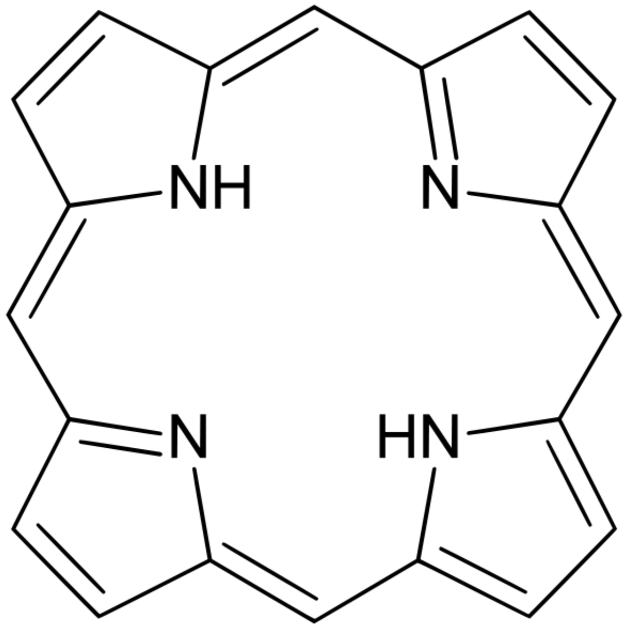



Database Center for Life Science (DBCLS)

- Important amino acid for **heme** synthesis
- All carbon and nitrogen from **glycine** or **succinyl CoA**



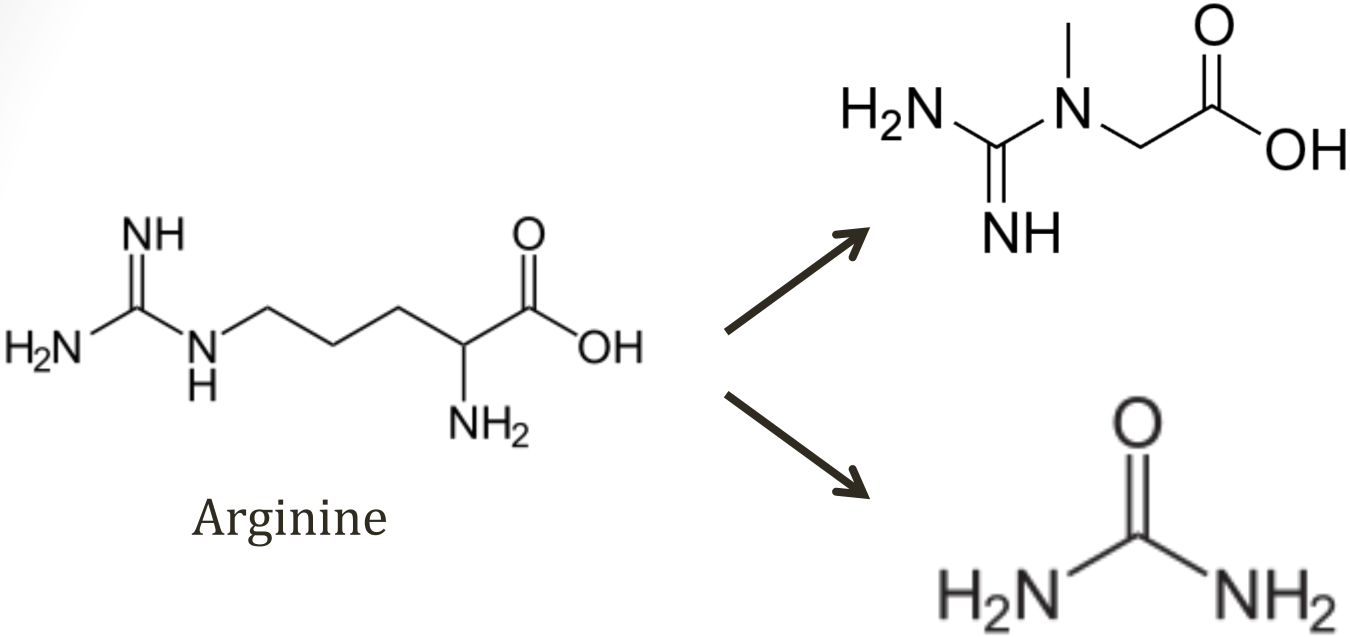

Heme




Porphyrin Ring

# Arginine

Creatine (muscle)



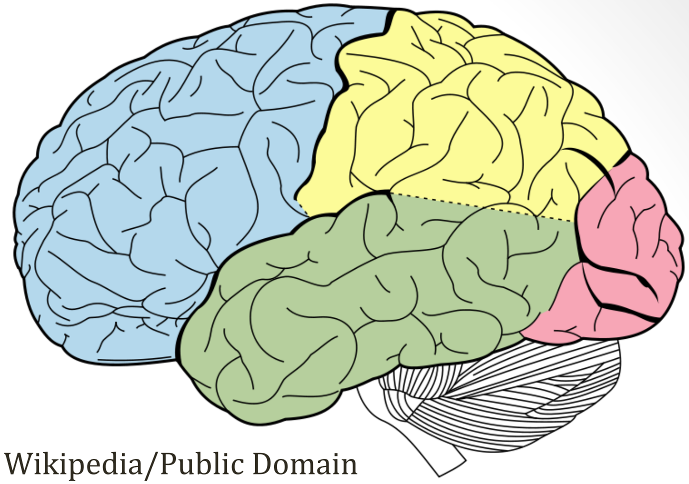
Arginine

Urea (urea cycle)

**Nitric Oxide Synthase**

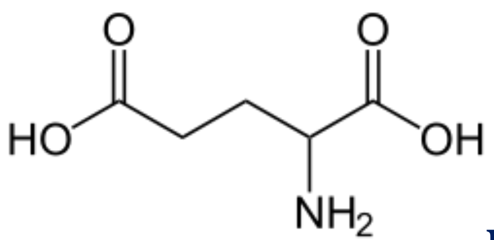
**Arginine** + NADPH → Citrulline + **Nitric Oxide** + NADP+

# Glutamate

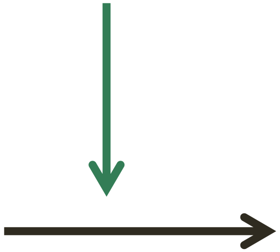


Wikipedia/Public Domain

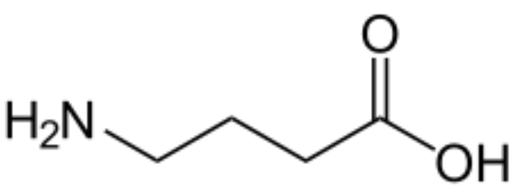
**Vitamin B6**




Glutamate



Glutamate Decarboxylase



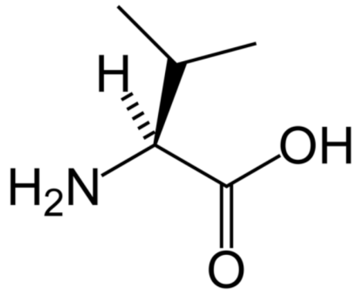

Gamma-aminobutyric acid
GABA

**Excitatory Neurotransmitter**

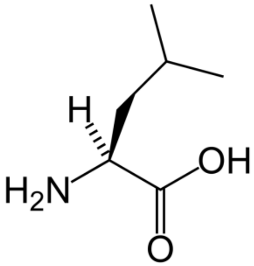
**Inhibitory Neurotransmitter**

# Branched Chain Amino Acids

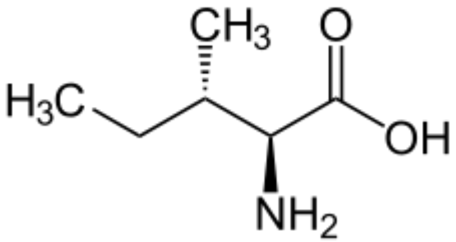
- Essential amino acids
- Primarily metabolized by muscle cells
- Metabolism depends on **α-ketoacid dehydrogenase**
  - Branched-chain α-ketoacid dehydrogenase complex (BCKDC)
  - Similar to pyruvate dehydrogenase complex
  - E1, E2, E3 subunits
  - Cofactors: Thiamine, lipoic acid



Valine

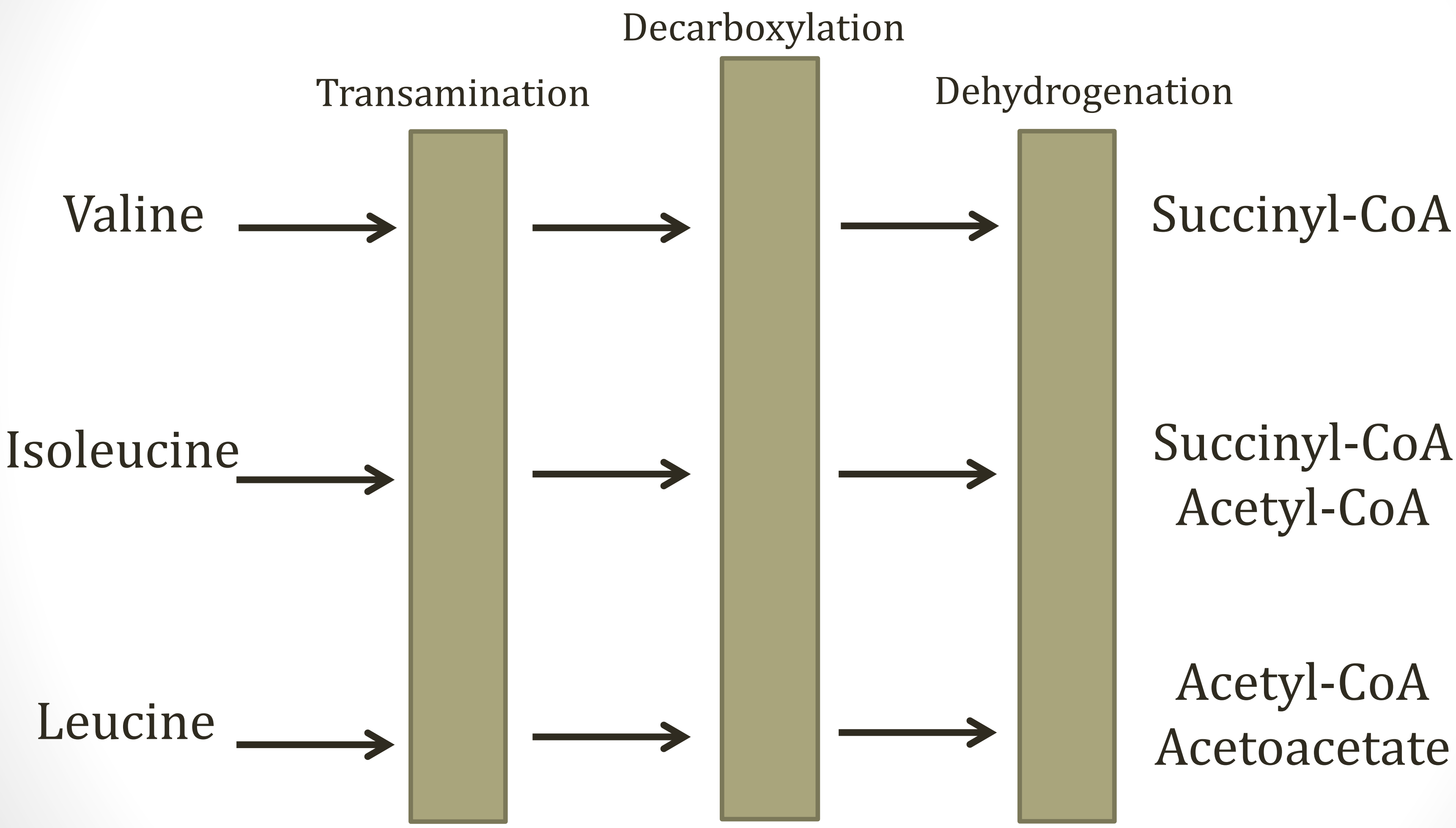


Leucine



Isoleucine

# Branched Chain Amino Acids



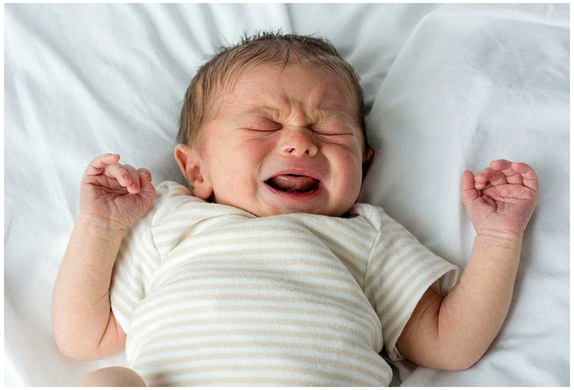

**α-ketoacid dehydrogenase**

# Maple Syrup Urine Disease

- Deficiency of **α-ketoacid dehydrogenase**
- Autosomal recessive
- Five phenotypes
- Classic MSUD most common (E1, E2, E3 deficiency)
- ↑ branched chain AA's and α-ketoacids in plasma
- α-ketoacid of isoleucine gives urine sweet smell

# Maple Syrup Urine Disease

- **Neurotoxicity** is main problem MSUD
- Primarily due to accumulation of **leucine**: "leucinosis"
- Classic MSUD occurs in 1$^{st}$ few days of life
- Lethargy and irritability
- Apnea, seizures
- Signs of cerebral edema



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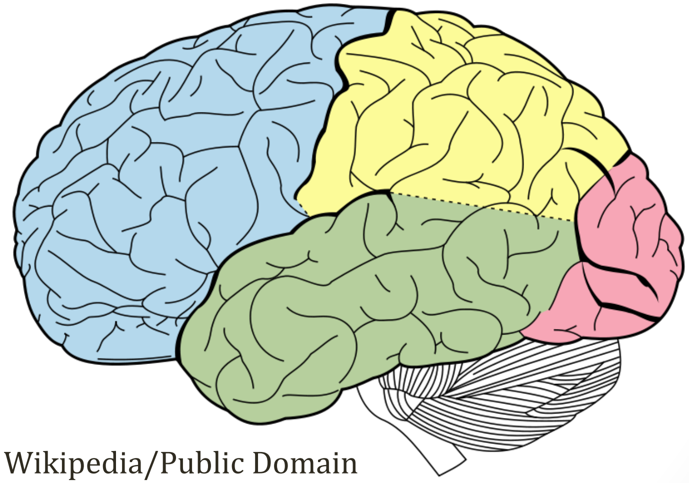

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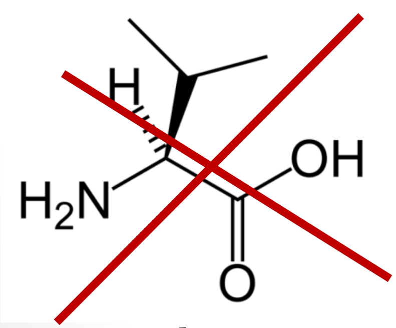
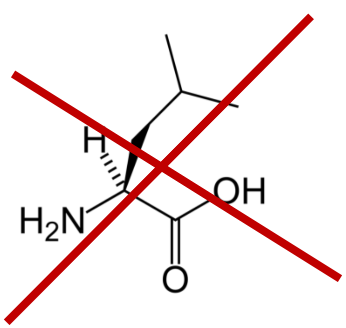
# Maple Syrup Urine Disease

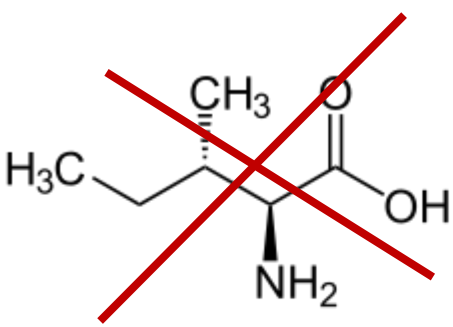
- Diagnosis:
  - Elevated **branched chain amino acid** levels in plasma
  - Valine, leucine, isoleucine
- Treatment:
  - **Dietary restriction** of branched-chain amino acids
  - Monitoring plasma amino acid concentrations
  - Thiamine supplementation




Valine



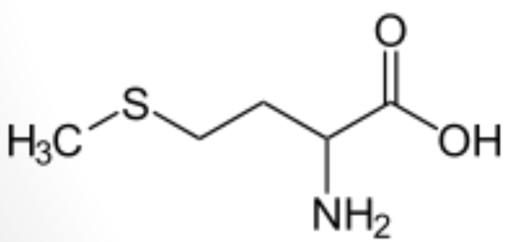

Leucine



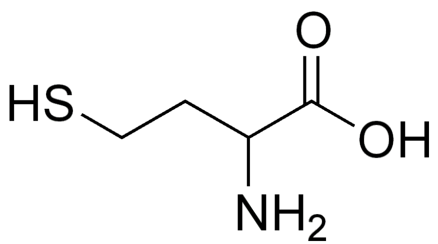

Isoleucine

# Homocysteine

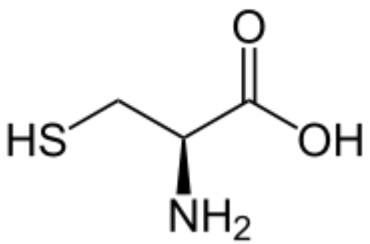
- **Homocysteine**, **cysteine**, and **methionine** related
- Methionine: essential
- Cysteine: non-essential
  - Synthesized from methionine
- Homocysteine: non-standard
- Transsulfuration pathway
  - Methionine → homocysteine → cysteine




**Methionine**

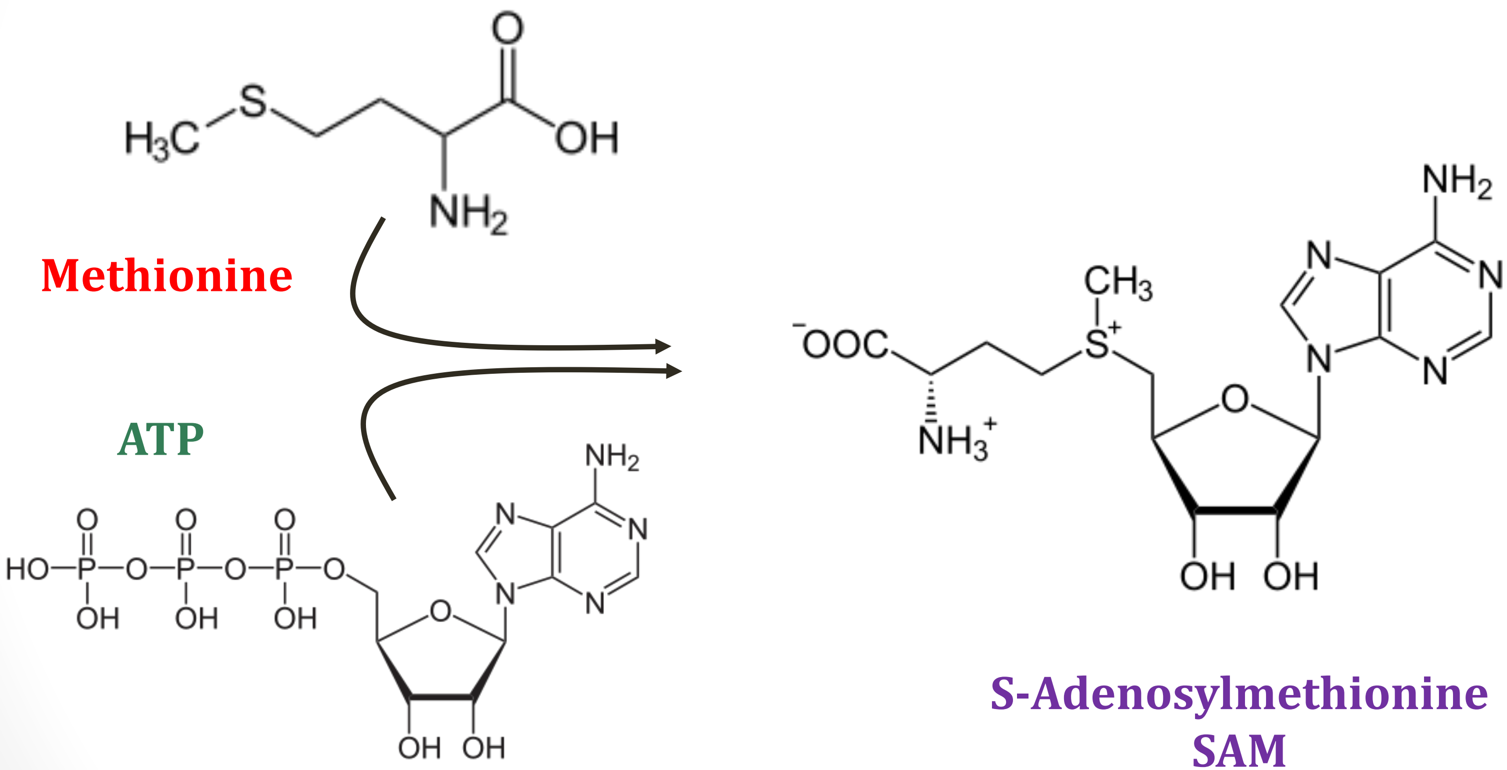



**Homocysteine**

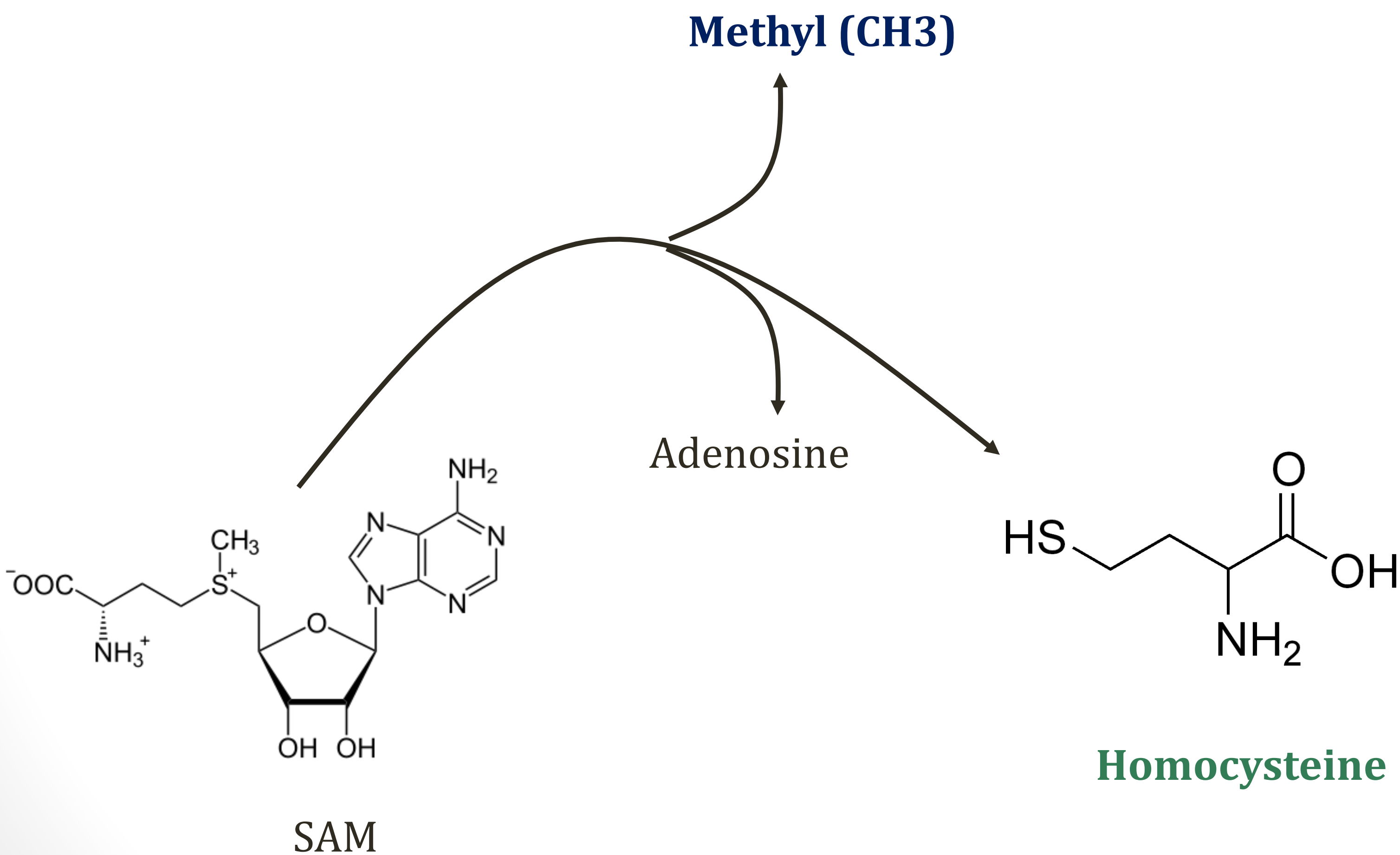



**Cysteine**

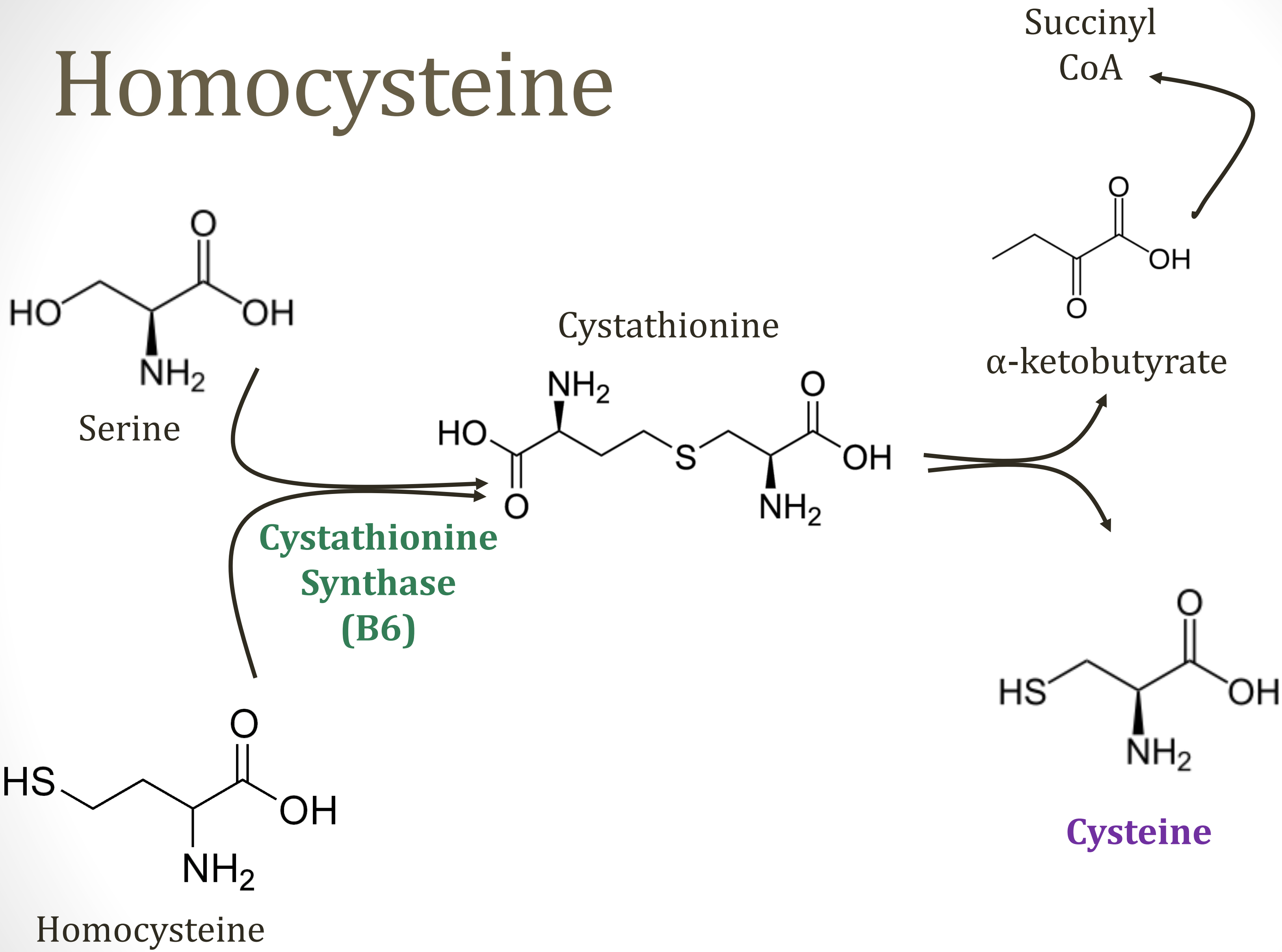
# Homocysteine



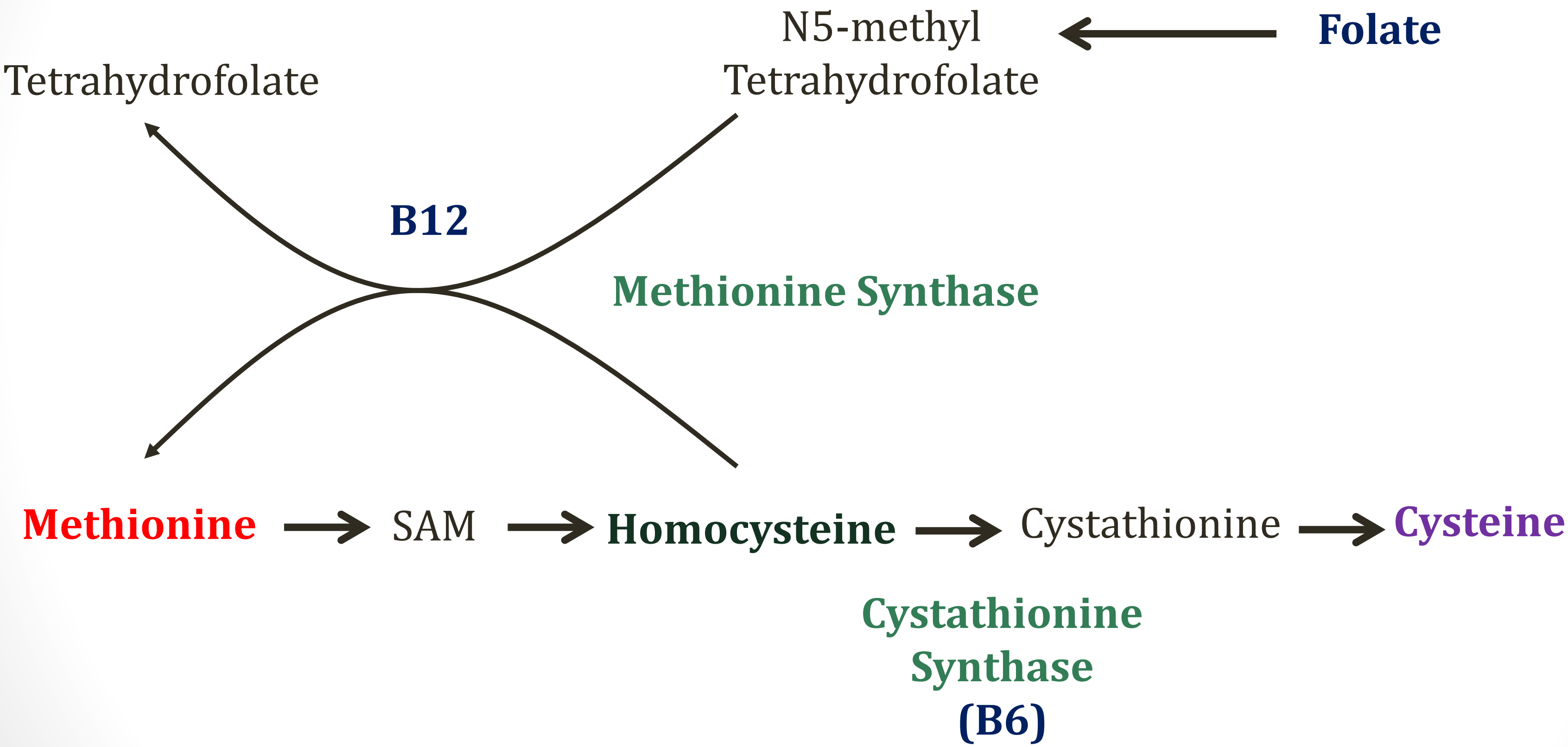
# Homocysteine



# Homocysteine



# Homocysteine



# Homocysteine Levels

- Normal: 5-15 micromoles/liter
- Mild-moderate elevations:
  - Can be caused by vitamin deficiencies: B12/folate, B6
  - May be associated with ↑ risk CV disease
  - No data on lowering levels to lower risk

# Homocystinuria

- Severe hyperhomocysteinemia: >100micromoles/liter
- Defects in homocysteine metabolism enzymes
- Autosomal recessive disorders

# Homocystinuria

- Common symptoms (mechanisms unclear)
  - **Lens** dislocation
  - Long limbs, chest deformities
  - Osteoporosis in childhood
  - Mental retardation
  - **Blood clots**
  - Early atherosclerosis (stroke, MI)

Ahellwig/Wikipedia

# Homocystinuria

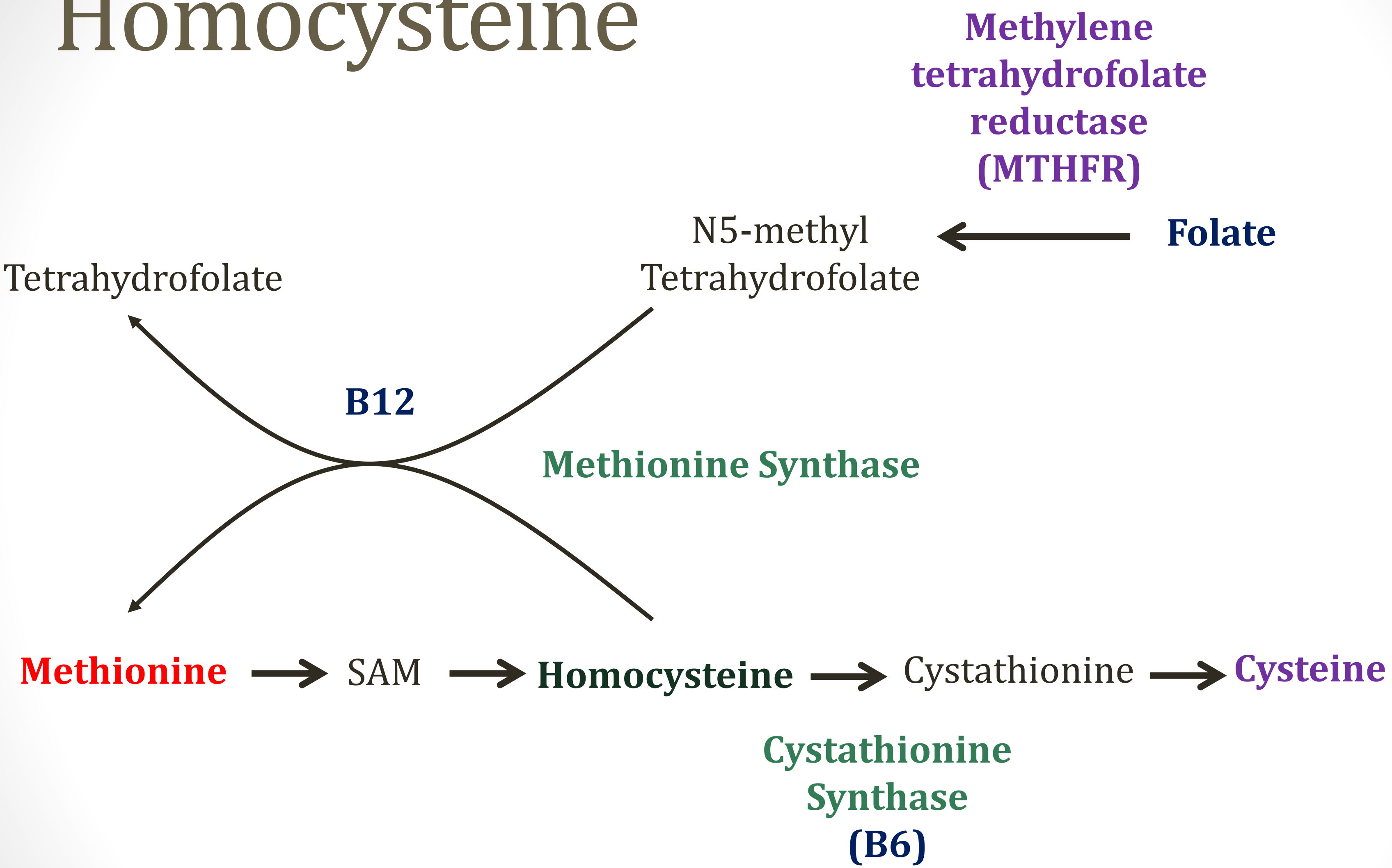
- Classic homocystinuria:
  - **Cystathionine β synthase (CBS)** deficiency
- Dietary treatment:
  - Avoid **methionine**
  - Increase cysteine (now essential)
  - Vitamin B6 supplementation (some patients "B6 responders")

# Homocysteine Elevations

## Less common causes

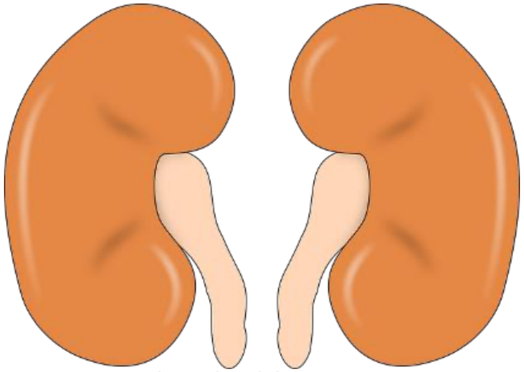
- Methionine synthase deficiency
- Defective metabolism folate/B12
  - MTHFR gene mutations

# Homocysteine



# Cystinuria

- Cystine: Two cysteine molecules linked together
- Cystinuria: autosomal recessive disorder
- ↓ reabsorption cystine by proximal tubule of nephron
- Main problem: **kidney stones**
- Prevention: **methionine** free diet



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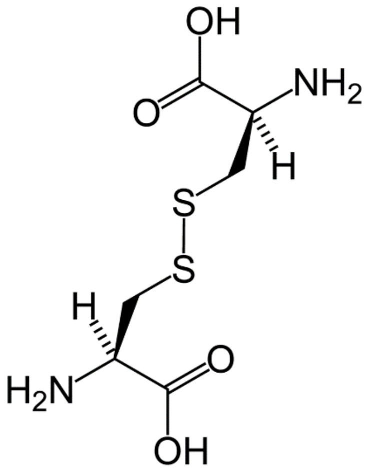


Cystine